

CHAPTER 4

MODEL ADEQUACY CHECKING

4.1 INTRODUCTION

The major **assumptions** that we have made thus far in our study of regression analysis are as follows:

1. The relationship between the response y and the regressors is linear, at least approximately.
2. The error term ε has zero mean.
3. The error term ε has constant variance σ^2 .
4. The errors are uncorrelated.
5. The errors are normally distributed.

Taken together, assumptions 4 and 5 imply that the errors are independent random variables. Assumption 5 is required for hypothesis testing and interval estimation.

We should always consider the validity of these assumptions to be doubtful and conduct analyses to examine the adequacy of the model we have tentatively entertained. The types of model inadequacies discussed here have potentially serious consequences. Gross violations of the assumptions may yield an unstable model in the sense that a different sample could lead to a totally different model with opposite conclusions. We usually cannot detect departures from the underlying assumptions by examination of the standard summary statistics, such as the t or F statistics, or R^2 . These are “global” model properties, and as such they do not ensure model adequacy.

In this chapter we present several methods useful for diagnosing violations of the basic regression assumptions. These diagnostic methods are primarily based on study of the model **residuals**. Methods for dealing with model inadequacies, as well as additional, more sophisticated diagnostics, are discussed in Chapters 5 and 6.

4.2 RESIDUAL ANALYSIS

4.2.1 Definition of Residuals

We have previously defined the residuals as

$$e_i = y_i - \hat{y}_i, \quad i = 1, 2, \dots, n \quad (4.1)$$

where y_i is an observation and $\hat{y}_i$ is the corresponding fitted value. Since a residual may be viewed as the **deviation** between the **data** and the **fit**, it is also a measure of the variability in the response variable not explained by the regression model. It is also convenient to think of the residuals as the realized or observed values of the model errors. Thus, any departures from the assumptions on the errors should show up in the residuals. Analysis of the residuals is an effective way to discover several types of model inadequacies. As we will see, **plotting residuals** is a very effective way to investigate how well the regression model fits the data and to check the assumptions listed in Section 4.1.

The residuals have several important properties. They have zero mean, and their approximate average variance is estimated by

$$\frac{\sum_{i=1}^n (e_i - \bar{e})^2}{n - p} = \frac{\sum_{i=1}^n e_i^2}{n - p} = \frac{SS_{\text{Res}}}{n - p} = MS_{\text{Res}}$$

The residuals are not independent, however, as the n residuals have only $n - p$ degrees of freedom associated with them. This nonindependence of the residuals has little effect on their use for model adequacy checking as long as n is not small relative to the number of parameters p .

4.2.2 Methods for Scaling Residuals

Sometimes it is useful to work with **scaled residuals**. In this section we introduce four popular methods for scaling residuals. These scaled residuals are helpful in finding observations that are **outliers**, or **extreme values**, that is, observations that are separated in some fashion from the rest of the data. See Figures 2.6–2.8 for examples of outliers and extreme values.

Standardized Residuals Since the approximate average variance of a residual is estimated by MS_{Res} , a logical scaling for the residuals would be the **standardized residuals**

$$d_i = \frac{e_i}{\sqrt{MS_{\text{Res}}}}, \quad i = 1, 2, \dots, n \quad (4.2)$$

The standardized residuals have mean zero and approximately unit variance. Consequently, a large standardized residual ($d_i > 3$, say) potentially indicates an outlier.

Studentized Residuals Using MS_{Res} as the variance of the i th residual, e_i is only an approximation. We can improve the residual scaling by dividing e_i by the exact standard deviation of the i th residual. Recall from Eq. (3.15b) that we may write the vector of residuals as

$$\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y} \quad (4.3)$$

where $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is the **hat matrix**. The hat matrix has several useful properties. It is **symmetric** ($\mathbf{H}' = \mathbf{H}$) and **idempotent** ($\mathbf{H}\mathbf{H} = \mathbf{H}$). Similarly the matrix $\mathbf{I} - \mathbf{H}$ is symmetric and idempotent. Substituting $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into Eq. (4.3) yields

$$\begin{aligned} \mathbf{e} &= (\mathbf{I} - \mathbf{H})(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \mathbf{X}\boldsymbol{\beta} - \mathbf{H}\mathbf{X}\boldsymbol{\beta} + (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon} \\ &= \mathbf{X}\boldsymbol{\beta} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\boldsymbol{\beta} + (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon} \end{aligned} \quad (4.4)$$

Thus, the residuals are the same linear transformation of the observations $\mathbf{y}$ and the errors $\boldsymbol{\varepsilon}$.

The covariance matrix of the residuals is

$$\text{Var}(\mathbf{e}) = \text{Var}[(\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}] = (\mathbf{I} - \mathbf{H})\text{Var}(\boldsymbol{\varepsilon})(\mathbf{I} - \mathbf{H})' = \sigma^2(\mathbf{I} - \mathbf{H}) \quad (4.5)$$

since $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$ and $\mathbf{I} - \mathbf{H}$ is symmetric and idempotent. The matrix $\mathbf{I} - \mathbf{H}$ is generally not diagonal, so the residuals have different variances and they are correlated.

The variance of the i th residual is

$$\text{Var}(e_i) = \sigma^2(1 - h_{ii}) \quad (4.6)$$

where h_{ii} is the i th diagonal element of the hat matrix $\mathbf{H}$. The covariance between residuals e_i and e_j is

$$\text{Cov}(e_i, e_j) = -\sigma^2 h_{ij} \quad (4.7)$$

where h_{ij} is the ij th element of the hat matrix. Now since $0 \leq h_{ii} \leq 1$, using the residual mean square MS_{Res} to estimate the variance of the residuals actually overestimates $\text{Var}(e_i)$. Furthermore, since h_{ii} is a measure of the **location** of the i th point in x space (recall the discussion of hidden extrapolation in Section 3.7), the variance of e_i depends on where the point $\mathbf{x}_i$ lies. Generally points near the center of the x space have larger variance (poorer least-squares fit) than residuals at more remote locations. Violations of model assumptions are more likely at remote points, and these violations may be hard to detect from inspection of the ordinary residuals e_i (or the standardized residuals d_i) because their residuals will usually be smaller.

Most students find it very counter-intuitive that the residuals for data points remote in terms of the x s are small, and in fact go to 0 as the remote points get

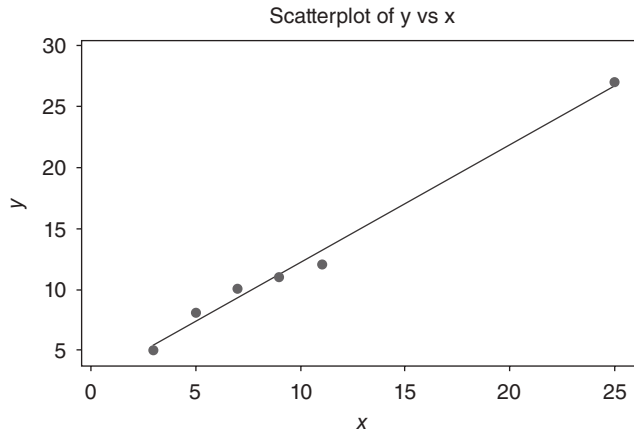


Figure 4.1 Example of a pure leverage point.

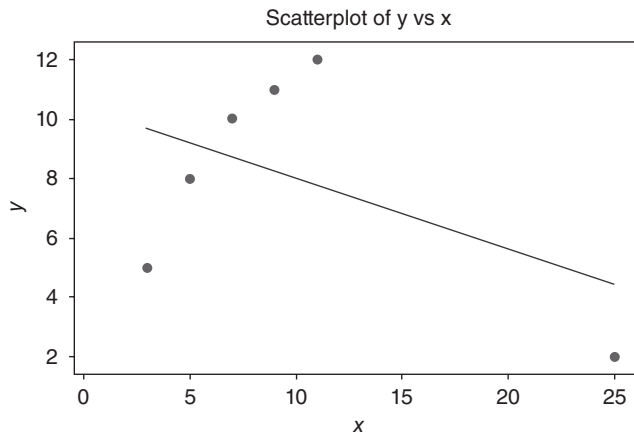


Figure 4.2 Example of an influential point.

further away from the center of the other points. Figures 4.1 and 4.2 help to illustrate this point. The only difference between these two plots occurs at $x = 25$. In Figure 4.1, the value of the response is 25, and in Figure 4.2, the value is 2. Figure 4.1 is a typical scatter plot for a pure **leverage** point. Such a point is remote in terms of the specific values of the regressors, but the observed value for the response is consistent with the prediction based on the other data values. The data point with $x = 25$ is an example of a pure leverage point. The line drawn on the figure is the actual ordinary least squares fit to the entire data set. Figure 4.2 is a typical scatter plot for an **influential** point. Such a data value is not only remote in terms of the specific values for the regressors, but the observed response is not consistent with the values that would be predicted based on only the other data points. Once again, the line drawn is the actual ordinary least squares fit to the entire data set. One can clearly see that the influential point draws the prediction equation to itself.

A little mathematics provides more insight into this situation. Let y_n be the observed response for the n^{th} data point, let x_n be the specific values for the regressors for this data point, let $\hat{y}_n^*$ be the predicted value for the response based on the other $n - 1$ data points, and let $\delta = y_n - \hat{y}_n^*$ be the difference between the actually observed value for this response compared to the predicted value based on the other values. Please note that $y_n = \hat{y}_n^* + \delta$. If a data point is remote in terms of the regressor values and $|\delta|$ is large, then we have an influential point. In Figures 4.1 and 4.2, consider $x = 25$. Let y_n be 2, the value from Figure 4.2. The actual predicted value for the that response based on the other four data values is 25, which is the point illustrated in Figure 4.1. In this case, $\delta = -23$, and we see that it is a very influential point. Finally, let $\hat{y}_n$ be the predicted value for the n^{th} response using all the data. It can be shown that

$$\hat{y}_n = \hat{y}_n^* + h_{nn}\delta$$

where h_{nn} is the n^{th} diagonal element of the hat matrix. If the n^{th} data point is remote in terms of the space defined by the data values for the regressors, then h_{nn} approaches 1, and $\hat{y}_n$ approaches y_n . The remote data value “drags” the prediction to itself.

This point is easier to see within a simple linear regression example. Let $\bar{x}^*$ be the average value for the other $n - 1$ regressors. It can be shown that

$$\hat{y}_n = \hat{y}_n^* + \left[\frac{1}{n} + \left(\frac{n-1}{n} \right)^2 \frac{(x_n - \bar{x}^*)^2}{S_{xx}} \right] \delta.$$

Clearly, for even a moderate sample size, as the data point becomes more remote in terms of the regressors (as x_n moves further away from $\bar{x}^*$, then the ordinary least squares estimate of y_n approaches the actually observed value for y_n).

The bottom line is two-fold. As we discussed in Sections 2.4 and 3.4, the prediction variance for data points that are remote in terms of the regressors is large. However, these data points do draw the prediction equation to themselves. As a result, the variance of the residuals for these points is small. This combination presents complications for doing proper residual analysis.

A logical procedure, then, is to examine the **studentized residuals**

$$r_i = \frac{e_i}{\sqrt{MS_{\text{Res}}(1 - h_{ii})}}, \quad i = 1, 2, \dots, n \quad (4.8)$$

instead of e_i (or d_i). The studentized residuals have constant variance $\text{Var}(r_i) = 1$ regardless of the location of $\mathbf{x}_i$ when the form of the model is correct. In many situations the variance of the residuals stabilizes, particularly for large data sets. In these cases there may be little difference between the standardized and studentized residuals. Thus, standardized and studentized residuals often convey equivalent information. However, since any point with a large residual **and** a large h_{ii} is potentially highly influential on the least-squares fit, examination of the studentized residuals is generally recommended.

Some of these points are very easy to see by examining the studentized residuals for a simple linear regression model. If there is only one regressor, it is easy to show that the studentized residuals are

$$r_i = \frac{e_i}{\sqrt{MS_{\text{Res}} \left[1 - \left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}} \right) \right]}}, \quad i = 1, 2, \dots, n \quad (4.9)$$

Notice that when the observation x_i is close to the midpoint of the x data, $x_i - \bar{x}$ will be small, and the estimated standard deviation of e_i [the denominator of Eq. (4.9)] will be large. Conversely, when x_i is near the extreme ends of the range of the x data, $x_i - \bar{x}$ will be large, and the estimated standard deviation of e_i will be small. Also, when the sample size n is really large, the effect of $(x_i - \bar{x})^2$ will be relatively small, so in big data sets, studentized residuals may not differ dramatically from standardized residuals.

PRESS Residuals The standardized and studentized residuals are effective in detecting outliers. Another approach to making residuals useful in finding outliers is to examine the quantity that is computed from $y_i - \hat{y}_{(i)}$, where $\hat{y}_{(i)}$ is the fitted value of the i th response based on all observations except the i th one. The logic behind this is that if the i th observation y_i is really unusual, the regression model based on all observations may be overly influenced by this observation. This could produce a fitted value $\hat{y}_i$ that is very similar to the observed value y_i , and consequently, the ordinary residual e_i will be small. Therefore, it will be hard to detect the outlier. However, if the i th observation is deleted, then $\hat{y}_{(i)}$ cannot be influenced by that observation, so the resulting residual should be likely to indicate the presence of the outlier.

If we delete the i th observation, fit the regression model to the remaining $n - 1$ observations, and calculate the predicted value of y_i corresponding to the deleted observation, the corresponding **prediction error** is

$$e_{(i)} = y_i - \hat{y}_{(i)} \quad (4.10)$$

This prediction error calculation is repeated for each observation $i = 1, 2, \dots, n$. These prediction errors are usually called **PRESS residuals** (because of their use in computing the prediction error sum of squares, discussed in Section 4.3). Some authors call the $e_{(i)}$ **deleted residuals**.

It would initially seem that calculating the PRESS residuals requires fitting n different regressions. However, it is possible to calculate PRESS residuals from the results of a single least-squares fit to all n observations. We show in Appendix C.7 how this is accomplished. It turns out that the i th PRESS residual is

$$e_{(i)} = \frac{e_i}{1 - h_{ii}}, \quad i = 1, 2, \dots, n \quad (4.11)$$

From Eq. (4.11) it is easy to see that the PRESS residual is just the ordinary residual weighted according to the diagonal elements of the hat matrix h_{ii} . Residuals associated with points for which h_{ii} is large will have large PRESS residuals. These points will generally be **high influence** points. Generally, a large difference between the ordinary residual and the PRESS residual will indicate a point where the model **fits** the data well, but a model built without that point **predicts** poorly. In Chapter 6, we discuss some other measures of influential observations.

Finally, the variance of the i th PRESS residual is

$$\text{Var}[e_{(i)}] = \text{Var}\left[\frac{e_i}{1-h_{ii}}\right] = \frac{1}{(1-h_{ii})^2}[\sigma^2(1-h_{ii})] = \frac{\sigma^2}{1-h_{ii}}$$

so that a **standardized** PRESS residual is

$$\frac{e_{(i)}}{\sqrt{\text{Var}[e_{(i)}]}} = \frac{e_i/(1-h_{ii})}{\sqrt{\sigma_i^2(1-h_{ii})}} = \frac{e_i}{\sqrt{\sigma^2(1-h_{ii})}}$$

which, if we use MS_{Res} to estimate σ^2 , is just the **studentized residual** discussed previously.

R-Student The studentized residual r_i discussed above is often considered an outlier diagnostic. It is customary to use MS_{Res} as an estimate of σ^2 in computing r_i . This is referred to as **internal scaling** of the residual because MS_{Res} is an internally generated estimate of σ^2 obtained from fitting the model to all n observations. Another approach would be to use an estimate of σ^2 based on a data set with the i th observation removed. Denote the estimate of σ^2 so obtained by $S_{(i)}^2$. We can show (see Appendix C.8) that

$$S_{(i)}^2 = \frac{(n-p)MS_{\text{Res}} - e_i^2/(1-h_{ii})}{n-p-1} \quad (4.12)$$

The estimate of σ^2 in Eq. (4.12) is used instead of MS_{Res} to produce an **externally studentized residual**, usually called **R-student**, given by

$$t_i = \frac{e_i}{\sqrt{S_{(i)}^2(1-h_{ii})}}, \quad i = 1, 2, \dots, n \quad (4.13)$$

In many situations t_i will differ little from the studentized residual r_i . However, if the i th observation is influential, then $S_{(i)}^2$ can differ significantly from MS_{Res} , and thus the *R*-student statistic will be more sensitive to this point.

It turns out that under the usual regression assumptions, t_i will follow the t_{n-p-1} distribution. Appendix C.9 establishes a formal hypothesis-testing procedure for outlier detection based on *R*-student. One could use a Bonferroni-type approach and compare all n values of $|t_i|$ to $t_{(\alpha/2n), n-p-1}$ to provide guidance regarding outliers. However, it is our view that a formal approach is usually not necessary and that only relatively crude cutoff values need be considered. In general, a **diagnostic view** as opposed to a strict statistical hypothesis-testing view is best. Furthermore, detection of outliers often needs to be considered simultaneously with detection of influential observations, as discussed in Chapter 6.

Example 4.1 The Delivery Time Data

Table 4.1 presents the scaled residuals discussed in this section using the model for the soft drink delivery time data developed in Example 3.1. Examining column

1 of Table 4.1 (the ordinary residuals, originally calculated in Table 3.3) we note that one residual, $e_9 = 7.4197$, seems suspiciously large. Column 2 shows that the standardized residual is $d_9 = e_9 / \sqrt{MS_{\text{Res}}} = 7.4197 / \sqrt{10.6239} = 2.2763$. All other standardized residuals are inside the ± 2 limits. Column 3 of Table 4.1 shows the studentized residuals. The studentized residual at point 9 is $r_9 = e_9 / \sqrt{MS_{\text{Res}}(1 - h_{9,9})} = 7.4197 / \sqrt{10.6239(1 - 0.49829)} = 3.2138$, which is substantially larger than the standardized residual. As we noted in Example 3.13, point 9 has the largest value of x_1 (30 cases) and x_2 (1460 feet). If we take the remote location of point 9 into account when scaling its residual, we conclude that the model does not fit this point well. The diagonal elements of the hat matrix, which are used extensively in computing scaled residuals, are shown in column 4.

Column 5 of Table 4.1 contains the PRESS residuals. The PRESS residuals for points 9 and 22 are substantially larger than the corresponding ordinary residuals, indicating that these are likely to be points where the model fits reasonably well but does not provide good predictions of fresh data. As we have observed in Example 3.13, these points are remote from the rest of the sample.

Column 6 displays the values of R -student. Only one value, t_9 , is unusually large. Note that t_9 is larger than the corresponding studentized residual r_9 , indicating that when run 9 is set aside, $S_{(9)}^2$ is smaller than MS_{Res} , so clearly this run is influential. Note that $S_{(9)}^2$ is calculated from Eq. (4.12) as follows:

$$\begin{aligned} S_{(9)}^2 &= \frac{(n-p)MS_{\text{Res}} - e_9^2 / (1 - h_{9,9})}{n-p-1} \\ &= \frac{(22)(10.6239) - (7.4197)^2 / (1 - 0.49829)}{21} \\ &= 5.9046 \end{aligned}$$

■

4.2.3 Residual Plots

As mentioned previously, graphical analysis of residuals is a very effective way to investigate the adequacy of the fit of a regression model and to check the underlying assumptions. In this section, we introduce and illustrate the basic residual plots. These plots are typically generated by regression computer software packages. They should be examined routinely in all regression modeling problems. We often plot externally studentized residuals because they have constant variance.

Normal Probability Plot Small departures from the normality assumption do not affect the model greatly, but gross nonnormality is potentially more serious as the t or F statistics and confidence and prediction intervals depend on the normality assumption. Furthermore, if the errors come from a distribution with thicker or heavier tails than the normal, the least-squares fit may be sensitive to a small subset of the data. Heavy-tailed error distributions often generate outlier that “pull” the least-squares fit too much in their direction. In these cases other estimation techniques (such as the **robust regression** methods in Section 15.1 should be considered.

A very simple method of checking the normality assumption is to construct a **normal probability** plot of the residuals. This is a graph designed so that the cumula-

TABLE 4.1 Scaled Residuals for Example 4.1

Observation	$e_i = y_i - \hat{y}_i$	$d_i = e_i / \sqrt{MS_{Res}}$	$r_i = e_i / \sqrt{MS_{Res}(1-h_{ii})}$	h_{ii}	$e_{(i)} = e_i / (1 - h_{ii})$	$t_i = e_i / \sqrt{S_{(i)}^2(1-h_{ii})}$	$[e_i / (1 - h_{ii})]^2$
Number, i	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1	-5.0281	-1.5426	-1.6277	0.10180	-5.5980	-1.6956	31.3373
2	1.1464	0.3517	0.349	0.07070	1.2336	0.3575	1.5218
3	-0.0498	-0.0153	-0.0161	0.09874	-0.0557	-0.0157	0.0031
4	4.9244	1.5108	1.5798	0.05838	5.2297	1.6392	27.3499
5	-0.4444	-0.1363	-0.1418	0.07501	-0.4804	-0.1386	0.2308
6	-0.2896	-0.0888	-0.0908	0.04287	-0.3025	-0.0887	0.0915
7	0.8446	0.2501	0.2704	0.08180	0.9198	0.2646	0.8461
8	1.1566	0.3548	0.3667	0.06373	1.2353	0.3594	1.5260
9	7.4197	2.2763	3.2138	0.49829	14.7888	4.3108	218.7093
10	2.3764	0.7291	0.8133	0.19630	2.9568	0.8068	8.728
11	2.2375	0.6865	0.7181	0.08613	2.4484	0.7099	5.9946
12	-0.5930	-0.1819	-0.1932	0.11366	-0.6690	-0.1890	0.4476
13	1.0270	0.3151	0.3252	0.06113	1.0938	0.3185	1.1965
14	1.0675	0.3275	0.3411	0.07824	1.1581	0.3342	1.3412
15	0.6712	0.2059	0.2103	0.04111	0.7000	0.2057	0.4900
16	-0.6629	-0.2034	-0.2227	0.16594	-0.7948	-0.2178	0.6317
17	0.4364	0.1339	0.1381	0.05943	0.4640	0.1349	0.2153
18	3.4486	1.0580	1.1130	0.09626	3.8159	1.1193	14.5612
19	1.7932	0.5502	0.5787	0.09645	1.9846	0.5698	3.9387
20	-5.7880	-1.7758	-1.8736	0.10169	-6.4432	-1.9967	41.5150
21	-2.6142	-0.8020	-0.8779	0.16528	-3.1318	-0.8731	9.8084
22	-3.6865	-1.1310	-1.4500	0.39158	-6.0591	-1.4896	36.7131
23	-4.6076	-1.4136	-1.4437	0.04126	-4.8059	-1.4825	23.0966
24	-4.5728	-1.4029	-1.4961	0.12061	-5.2000	-1.5422	27.0397
25	-0.2126	-0.0652	-0.0675	0.06664	-0.2278	-0.0660	0.0519
							PRESS = 457.4000

tive normal distribution will plot as a straight line. Let $t_{[1]} < t_{[2]} < \dots < t_{[n]}$ be the externally studentized residuals ranked in increasing order. If we plot $t_{[i]}$ against the cumulative probability $P_i = (i - \frac{1}{2})/n$, $i = 1, 2, \dots, n$, on the normal probability plot, the resulting points should lie approximately on a straight line. The straight line is usually determined visually, with emphasis on the central values (e.g., the 0.33 and 0.67 cumulative probability points) rather than the extremes. Substantial departures from a straight line indicate that the distribution is not normal. Sometimes normal probability plots are constructed by plotting the ranked residual $t_{[i]}$ against the “expected normal value” $\Phi^{-1}[(i - \frac{1}{2})/n]$, where Φ denotes the standard normal cumulative distribution. This follows from the fact that $E(t_{[i]}) \approx \Phi^{-1}[(i - \frac{1}{2})/n]$.

Figure 4.3a displays an “idealized” normal probability plot. Notice that the points lie approximately along a straight line. Panels b–e present other typical problems. Panel b shows a sharp upward and downward curve at both extremes, indicating that the tails of this distribution are too light for it to be considered normal. Conversely, panel c shows flattening at the extremes, which is a pattern typical of samples from a distribution with heavier tails than the normal. Panels d and e exhibit patterns associated with positive and negative skew, respectively.[†]

Because samples taken from a normal distribution will not plot exactly as a straight line, some experience is required to interpret normal probability plots. Daniel and Wood [1980] present normal probability plots for sample sizes 8–384. Study of these plots is helpful in acquiring a feel for how much deviation from the

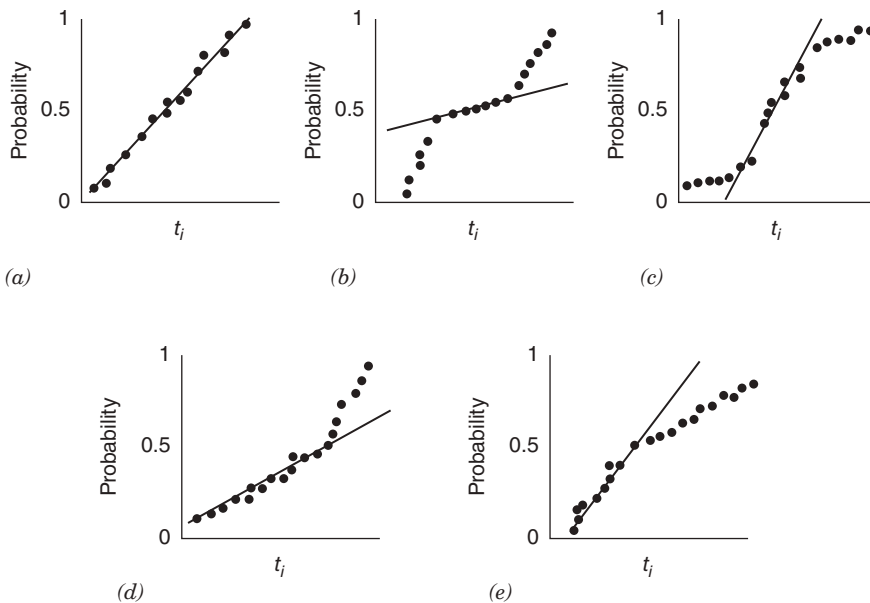


Figure 4.3 Normal probability plots: (a) ideal; (b) light-tailed distribution; (c) heavy-tailed distribution; (d) positive skew; (e) negative skew.

[†]These interpretations assume that the ranked residuals are plotted on the horizontal axis. If the residuals are plotted on the vertical axis, as some computer systems do, the interpretation is reversed.

straight line is acceptable. Small sample sizes ($n \leq 16$) often produce normal probability plots that deviate substantially from linearity. For larger sample sizes ($n \geq 32$) the plots are much better behaved. Usually about 20 points are required to produce normal probability plots that are stable enough to be easily interpreted.

Andrews [1979] and Gnanadesikan [1977] note that normal probability plots often exhibit no unusual behavior even if the errors ε_i are not normally distributed. This problem occurs because the residuals are not a simple random sample; they are the remnants of a parameter estimation process. The residuals are actually linear combinations of the model errors (the ε_i). Thus, fitting the parameters tends to destroy the evidence of nonnormality in the residuals, and consequently we cannot always rely on the normal probability plot to detect departures from normality.

A common defect that shows up on the normal probability plot is the occurrence of one or two large residuals. Sometimes this is an indication that the corresponding observations are **outliers**. For additional discussion of outliers, refer to Section 4.4.

Example 4.2 The Delivery Time Data

Figure 4.4 presents a normal probability plot of the externally studentized residuals from the regression model for the delivery time data from Example 3.1. The residuals are shown in columns 1 and 2 of Table 4.1.

The residuals do not lie exactly along a straight line, indicating that there may be some problems with the normality assumption, or that there may be one or more outliers in the data. From Example 4.1, we know that the studentized residual for observation 9 is moderately large ($r_9 = 3.2138$), as is the R -student residual ($t_9 = 4.3108$). However, there is no indication of a severe problem in the delivery time data. ■

Plot of Residuals against the Fitted Values $\hat{y}_i$ A plot of the (preferably the externally studentized residuals, t_i) versus the corresponding fitted values $\hat{y}_i$ is useful for detecting several common types of model inadequacies.[†] If this plot resembles Figure 4.5a, which indicates that the residuals can be contained in a horizontal band, then there are no obvious model defects. Plots of t_i versus $\hat{y}_i$ that resemble any of the patterns in panels *b–d* are symptomatic of model deficiencies.

The patterns in panels *b* and *c* indicate that the variance of the errors is not constant. The **outward-opening funnel pattern** in panel *b* implies that the variance is an increasing function of y [an inward-opening funnel is also possible, indicating that $\text{Var}(\varepsilon)$ increases as y decreases]. The double-bow pattern in panel *c* often occurs when y is a proportion between zero and 1. The variance of a binomial proportion near 0.5 is greater than one near zero or 1. The usual approach for dealing with inequality of variance is to apply a suitable **transformation** to either the regressor or the response variable (see Sections 5.2 and 5.3) or to use the method of weighted least squares (Section 5.5). In practice, transformations on the response are generally employed to stabilize variance.

A curved plot such as in panel *d* indicates **nonlinearity**. This could mean that other regressor variables are needed in the model. For example, a squared term may

[†]The residuals should be plotted versus the fitted values $\hat{y}_i$ and not the observed values y_i because the e_i and the $\hat{y}_i$ are uncorrelated while the e_i and the y_i are usually correlated. The proof of this statement in Appendix C.10.

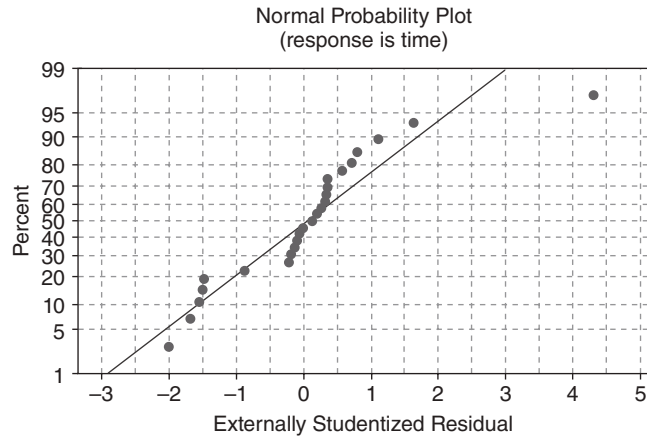


Figure 4.4 Normal probability plot of the externally studentized residuals for the delivery time data.

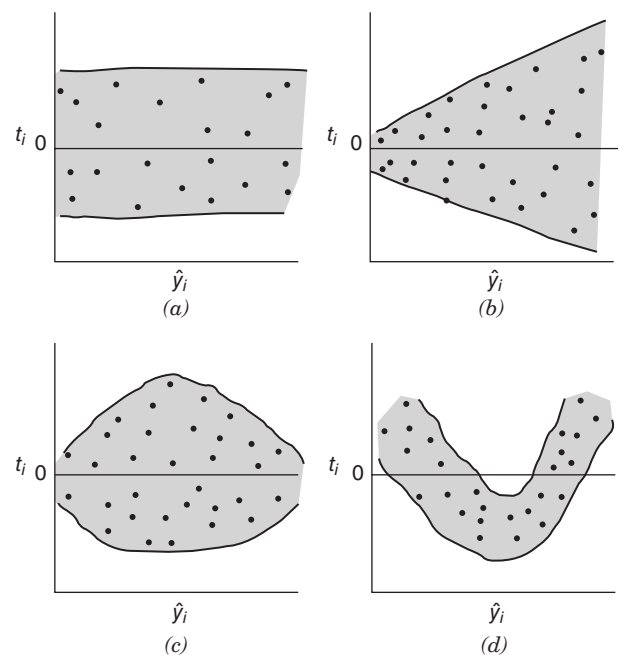


Figure 4.5 Patterns for residual plots: (a) satisfactory; (b) funnel; (c) double bow; (d) nonlinear.

be necessary. Transformations on the regressor and/or the response variable may also be helpful in these cases.

A plot of the residuals against $\hat{y}_i$ may also reveal one or more unusually large residuals. These points are, of course, potential outliers. Large residuals that occur at the extreme $\hat{y}_i$ values could also indicate that either the variance is not constant or the true relationship between y and x is not linear. These possibilities should be investigated before the points are considered outliers.

Example 4.3 The Delivery Time Data

Figure 4.6 presents the plot of the externally studentized residuals versus the fitted values of delivery time. The plot does not exhibit any strong unusual pattern, although the large residual t_9 shows up clearly. There does seem to be a slight tendency for the model to underpredict short delivery times and overpredict long delivery times.

Plot of Residuals against the Regressor Plotting the residuals against the corresponding values of each regressor variable can also be helpful. These plots often exhibit patterns such as those in Figure 4.5, except that the horizontal scale is x_{ij} for the j th regressor rather than $\hat{y}_i$. Once again an impression of a horizontal band containing the residuals is desirable. The funnel and double-bow patterns in panels *b* and *c* indicate nonconstant variance. The curved band in panel *d* or a nonlinear pattern in general implies that the assumed relationship between y and the regressor x_j is not correct. Thus, either higher order terms in x_j (such as x_j^2) or a transformation should be considered.

In the simple linear regressor case, it is not necessary to plot residuals versus both $\hat{y}_i$ and the regressor variable. The reason is that the fitted values $\hat{y}_i$ are linear combinations of the regressor values x_i , so the plots would only differ in the scale for the abscissa.

Example 4.4 The Delivery Time Data

Figure 4.7 presents the plots of the externally studentized residuals t_i from the delivery time problem in Example 3.1 versus both regressors. Panel *a* plots residuals versus cases and panel *b* plots residuals versus distance. Neither of these plots reveals any clear indication of a problem with either misspecification of the regressor (implying the need for either a transformation on the regressor or higher order

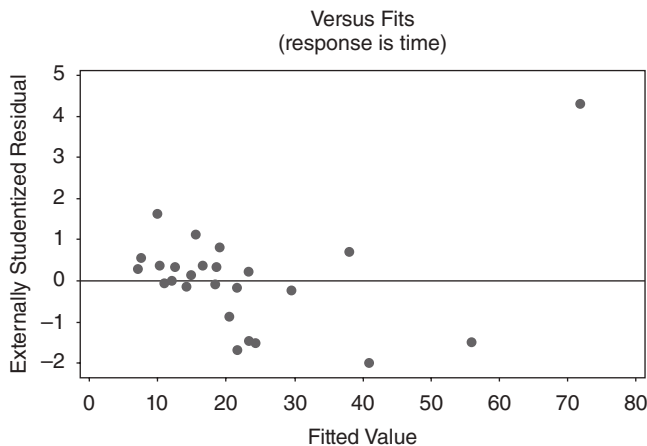


Figure 4.6 Plot of externally studentized residuals versus predicted for the delivery time data.

terms in cases and/or distance) or inequality of variance, although the moderately large residual associated with point 9 is apparent on both plots.

It is also helpful to plot residuals against regressor variables that are **not currently in the model** but which could potentially be included. Any structure in the plot of residuals versus an omitted variable indicates that incorporation of that variable could improve the model.

Plotting residuals versus a regressor is not always the most effective way to reveal whether a curvature effect (or a transformation) is required for that variable in the model. In Section 4.2.4 we describe two additional residual plots that are more effective in investigating the relationship between the response variable and the regressors. ■

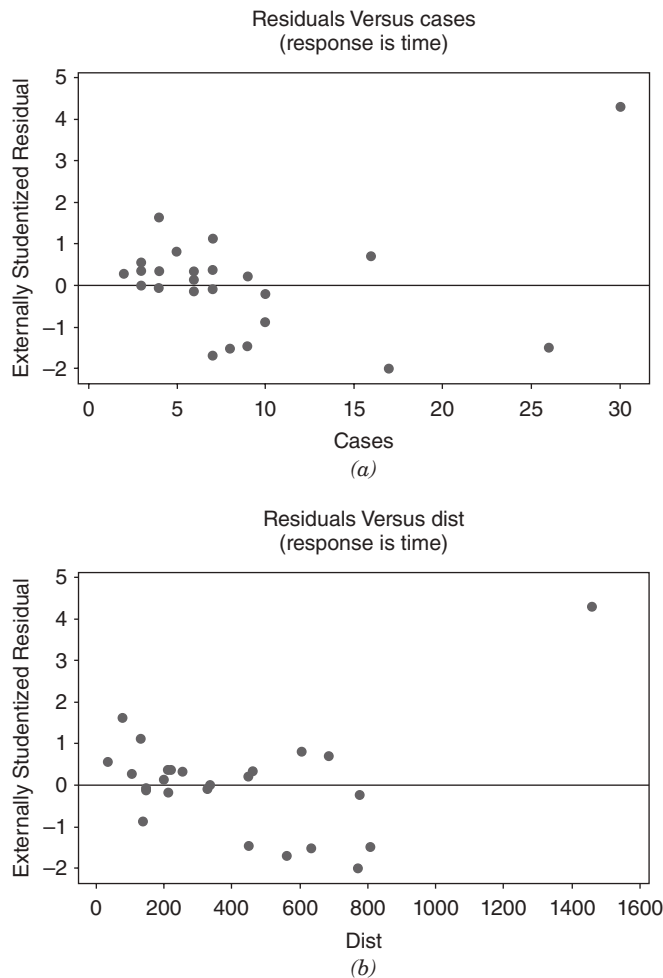


Figure 4.7 Plot of externally studentized residuals versus the regressors for the delivery time data: (a) residuals versus cases; (b) residuals versus distance.

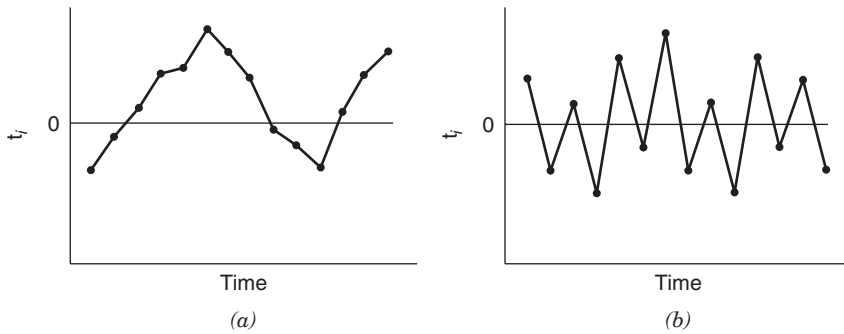


Figure 4.8 Prototype residual plots against time displaying autocorrelation in the errors: (a) positive autocorrelation; (b) negative autocorrelation.

Plot of Residuals in Time Sequence If the time sequence in which the data were collected is known, it is a good idea to plot the residuals against time order. Ideally, this plot will resemble Figure 4.5a; that is, a horizontal band will enclose all of the residuals, and the residuals will fluctuate in a more or less random fashion within this band. However, if this plot resembles the patterns in Figures 4.5b–d, this may indicate that the variance is changing with time or that linear or quadratic terms in time should be added to the model.

The time sequence plot of residuals may indicate that the errors at one time period are correlated with those at other time periods. The correlation between model errors at different time periods is called **autocorrelation**. A plot such as Figure 4.8a indicates positive autocorrelation, while Figure 4.8b is typical of negative autocorrelation. The presence of autocorrelation is a potentially serious violation of the basic regression assumptions. More discussion about methods for detecting autocorrelation and remedial measures are discussed in Chapter 14.

4.2.4 Partial Regression and Partial Residual Plots

We noted in Section 4.2.3 that a plot of residuals versus a regressor variable is useful in determining whether a curvature effect for that regressor is needed in the model. A limitation of these plots is that they may not completely show the correct or complete marginal effect of a regressor, given the other regressors in the model. A **partial regression plot** is a variation of the plot of residuals versus the predictor that is an enhanced way to study the marginal relationship of a regressor given the other variables that are in the model. This plot can be very useful in evaluating whether we have specified the relationship between the response and the regressor variables correctly. Sometimes the partial residual plot is called the **added-variable plot** or the **adjusted-variable plot**. Partial regression plots can also be used to provide information about the marginal usefulness of a variable that is not currently in the model.

Partial regression plots consider the marginal role of the regressor x_j given other regressors that are already in the model. In this plot, the response variable y and the regressor x_j are both regressed against the other regressors in the model and the residuals obtained for each regression. The plot of these residuals against each other provides information about the nature of the marginal relationship for regressor x_j under consideration.

To illustrate, suppose we are considering a first-order multiple regression model with two regressors variables, that is, $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$. We are concerned about the nature of the marginal relationship for regressor x_1 —in other words, is the relationship between y and x_1 correctly specified? First we would regress y on x_2 and obtain the fitted values and residuals:

$$\begin{aligned}\hat{y}_i(x_2) &= \hat{\theta}_0 + \hat{\theta}_1 x_{i2} \\ e_i(y|x_2) &= y_i - \hat{y}_i(x_2), \quad i = 1, 2, \dots, n\end{aligned}\quad (4.14)$$

Now regress x_1 on x_2 and calculate the residuals:

$$\begin{aligned}\hat{x}_{i1}(x_2) &= \hat{\alpha}_0 + \hat{\alpha}_1 x_{i2} \\ e_i(x_1|x_2) &= x_{i1} - \hat{x}_{i1}(x_2), \quad i = 1, 2, \dots, n\end{aligned}\quad (4.15)$$

The partial regression plot for regressor variable x_1 is obtained by plotting the y residuals $e_i(y|x_2)$ against the x_1 residuals $e_i(x_1|x_2)$. If the regressor x_1 enters the model linearly, then the partial regression plot should show a linear relationship, that is, the partial residuals will fall along a straight line with a nonzero slope. The slope of this line will be the regression coefficient of x_1 in the multiple linear regression model. If the partial regression plot shows a curvilinear band, then higher order terms in x_1 or a transformation (such as replacing x_1 with $1/x_1$) may be helpful. When x_1 is a **candidate** variable being considered for inclusion in the model, a horizontal band on the partial regression plot indicates that there is no additional useful information in x_1 for predicting y .

Example 4.5 The Delivery Time Data

Figure 4.9 presents the partial regression plots for the delivery time data, with the plot for x_1 shown in Figure 4.9a and the plot for x_2 shown in Figure 4.9b. The linear relationship between both cases and distance is clearly evident in both of these plots, although, once again, observation 9 falls somewhat off the straight line that apparently well-describes the rest of the data. This is another indication that point 9 bears further investigation. ■

Some Comments on Partial Regression Plots

1. Partial regression plots need to be used with caution as they only suggest **possible** relationships between the regressor and the response. These plots may not give information about the proper form of the relationship if several variables already in the model are incorrectly specified. It will usually be necessary to investigate several alternate forms for the relationship between the regressor and y or several transformations. Residual plots for these subsequent models should be examined to identify the best relationship or transformation.
2. Partial regression plots will not, in general, detect **interaction** effects among the regressors.

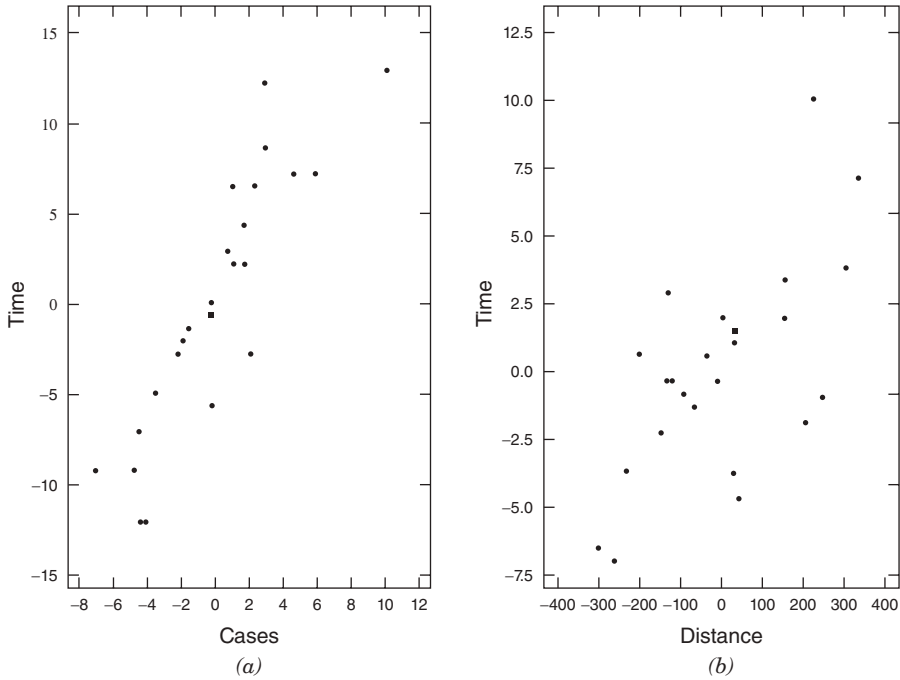


Figure 4.9 Partial regression plots for the delivery time data.

3. The presence of strong multicollinearity (refer to Section 3.9 and Chapter 9) can cause partial regression plots to give incorrect information about the relationship between the response and the regressor variables.
4. It is fairly easy to give a general development of the partial regression plotting concept that shows clearly why the slope of the plot should be the regression coefficient for the variable of interest, say x_j .

The partial regression plot is a plot of residuals from which the linear dependence of y on all regressors other than x_j has been removed against regressor x_j with its linear dependence on other regressors removed. In matrix form, we may write these quantities as $\mathbf{e}[y|\mathbf{X}_{(j)}]$ and $\mathbf{e}[x_j|\mathbf{X}_{(j)}]$, respectively, where $\mathbf{X}_{(j)}$ is the original $\mathbf{X}$ matrix with the j th regressor (x_j) removed. To show how these quantities are defined, consider the model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \mathbf{X}_{(j)}\boldsymbol{\beta} + \beta_j\mathbf{x}_j + \boldsymbol{\varepsilon} \quad (4.16)$$

Premultiply Eq. (4.16) by $\mathbf{I} - \mathbf{H}_{(j)}$ to give

$$(\mathbf{I} - \mathbf{H}_{(j)})\mathbf{y} = (\mathbf{I} - \mathbf{H}_{(j)})\mathbf{X}_{(j)}\boldsymbol{\beta} + \beta_j(\mathbf{I} - \mathbf{H}_{(j)})\mathbf{x}_j + (\mathbf{I} - \mathbf{H}_{(j)})\boldsymbol{\varepsilon}$$

and note that $(\mathbf{I} - \mathbf{H}_{(j)})\mathbf{X}_{(j)} = \mathbf{0}$, so that

$$(\mathbf{I} - \mathbf{H}_{(j)})\mathbf{y} = \beta_j(\mathbf{I} - \mathbf{H}_{(j)})\mathbf{x}_j + (\mathbf{I} - \mathbf{H}_{(j)})\boldsymbol{\varepsilon}$$

or

$$\mathbf{e}[\mathbf{y} | \mathbf{X}_{(j)}] = \beta_j \mathbf{e}[\mathbf{x}_j | \mathbf{X}_{(j)}] + \boldsymbol{\varepsilon}^*$$

where $\boldsymbol{\varepsilon}^* = (\mathbf{I} - \mathbf{H}_{(j)})\boldsymbol{\varepsilon}$. This suggests that a partial regression plot should have slope β_j . Thus, if x_j enters the regression in a linear fashion, the partial regression plot should show a linear relationship passing through the origin. Many computer programs (such as SAS and Minitab) will generate partial regression plots.

Partial Residual Plots A residual plot closely related to the partial regression plot is the **partial residual plot**. It is also designed to show the relationship between the response variable and the regressors. Suppose that the model contains the regressors $x_1, x_2, \dots, x_k$. The **partial residuals** for regressor x_j are defined as

$$e_i^*(y|x_j) = e_i + \hat{\beta}_j x_{ij}, \quad i = 1, 2, \dots, n$$

where the e_i are the residuals from the model with all k regressors included. When the partial residuals are plotted against x_{ij} , the resulting display has slope $\hat{\beta}_j$, the regression coefficient associated with x_j in the model. The interpretation of the partial residual plot is very similar to that of the partial regression plot. See Larsen and McCarty [1972], Daniel and Wood [1980], Wood [1973], Mallows [1986], Mansfield and Conerly [1987], and Cook [1993] for more details and examples.

4.2.5 Using Minitab®, SAS, and R for Residual Analysis

It is easy to generate the residual plots in Minitab. Select the “graphs” box. Once it is opened, select the “deleted” option to get the studentized residuals. You then select the residual plots you want.

Table 4.2 gives the SAS source code for SAS version 9 to do residual analysis for the delivery time data. The partial option provides the partial regression plots. A common complaint about SAS is the quality of many of the plots generated by its procedures. These partial regression plots are prime examples. Version 9, however, upgrades some of the more important graphics plots for PROC REG. The first plot statement generates the studentized residuals versus predicted values, the studentized residuals versus the regressors, and the studentized residuals by time plots (assuming that the order in which the data are given is the actual time order). The second plot statement gives the normal probability plot of the studentized residuals.

As we noted, over the years the basic plots generated by SAS have been improved. Table 4.3 gives appropriate source code for earlier versions of SAS that produce “nice” residual plots. This code is important when we discuss plots from other SAS procedures that still do not generate nice plots. Basically this code uses the OUTPUT command to create a new data set that includes all of the previous delivery information plus the predicted values and the studentized residuals. It then uses the SAS-GRAPH features of SAS to generate the residual plots. The code uses PROC CAPABILITY to generate the normal probability plot. Unfortunately, PROC CAPABILITY by default produces a lot of noninteresting information in the output file.

TABLE 4.2 SAS Code for Residual Analysis of Delivery Time Data

```

date delivery;
input time cases distance;
cards;
16.68 7 560
11.50 3 220
12.03 3 340
14.88 4 80
13.75 6 150
18.11 7 330
 8.00 2 110
17.83 7 210
79.24 30 1460
21.50 5 605
40.33 16 688
21.00 10 215
13.50 4 255
19.75 6 462
24.00 9 448
29.00 10 776
15.35 6 200
19.00 7 132
 9.50 3 36
35.10 17 770
17.90 10 140
52.32 26 810
18.75 9 450
19.83 8 635
10.75 4 150
proc reg;
  model time = cases distance / partial;
  plot rstudent.*(predicted. cases distance obs.);
  plot npp.*rstudent.;
run;

```

We next illustrate how to use R to create appropriate residual plots. Once again, consider the delivery data. The first step is to create a space delimited file named `delivery.txt`. The names of the columns should be `time`, `cases`, and `distance`.

The R code to do the basic analysis and to create the appropriate residual plots based on the externally studentized residuals is:

```

deliver <- read.table("delivery.txt",header=TRUE, sep=" ")
deliver.model <- lm(time~cases+distance, data=deliver)
summary(deliver.model)
yhat <- deliver.model$fit
t <- rstudent(deliver.model)
qqnorm(t)
plot(yhat,t)
plot(deliver$x1,t)
plot(deliver$x2,t)

```

TABLE 4.3 Older SAS Code for Residual Analysis of Delivery Time Data

```

date delivery;
input time cases distance;
cards;
16.68 7 560
11.50 3 220
12.03 3 340
14.88 4 80
13.75 6 150
18.11 7 330
 8.00 2 110
17.83 7 210
79.24 30 1460
21.50 5 605
40.33 16 688
21.00 10 215
13.50 4 255
19.75 6 462
24.00 9 448
29.00 10 776
15.35 6 200
19.00 7 132
 9.50 3 36
35.10 17 770
17.90 10 140
52.32 26 810
18.75 9 450
19.83 8 635
10.75 4 150
proc reg;
  model time = cases distance / partial;
  output out = delivery2 p = ptime rstudent= t;
run;
data delivery3;
  set delivery2;
  index = _n_;
proc gplot data = delivery3;
  plot t*ptime t*cases t*distance t*index;
run;
proc capability data = delivery3;
  var t;
  qqplot t;
run;

```

Generally, the graphics in R require a great deal of work in order to be of suitable quality. The commands

```
deliver2 <- cbind(deliver,yhat,t)
write.table(deliver2,"delivery_output.txt")
```

create a file “delivery_output.txt” which the user then can import into his/her favorite package for doing graphics.

4.2.6 Other Residual Plotting and Analysis Methods

In addition to the basic residual plots discussed in Sections 4.2.3 and 4.2.4, there are several others that are occasionally useful. For example, it may be very useful to construct a scatterplot of regressor x_i against regressor x_j . This plot may be useful in studying the relationship between regressor variables and the disposition of the data in x space. Consider the plot of x_i versus x_j in Figure 4.10. This display indicates that x_i and x_j are highly positively correlated. Consequently, it may not be necessary to include both regressors in the model. If two or more regressors are highly correlated, it is possible that multicollinearity is present in the data. As observed in Chapter 3 (Section 3.10), multicollinearity can seriously disturb the least-squares fit and in some situations render the regression model almost useless. Plots of x_i versus x_j may also be useful in discovering points that are remote from the rest of the data and that potentially influence key model properties. Anscombe [1973] presents several other types of plots between regressors. Cook and Weisberg [1994] give a very modern treatment of regression graphics, including many advanced techniques not considered in this book.

Figure 4.11 is a scatterplot of x_1 (cases) versus x_2 (distance) for delivery time data from Example 3.1 (Table 3.2). Comparing Figure 4.11 with Figure 4.10, we see that cases and distance are positively correlated. In fact, the simple correlation between x_1 and x_2 is $r_{12} = 0.82$. While highly correlated regressors can cause a number of serious problems in regression, there is no strong indication in this example that

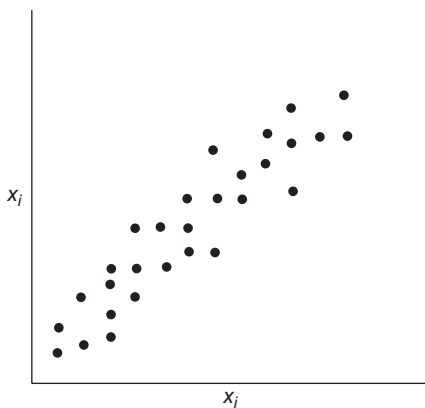


Figure 4.10 Plot of x_i versus x_j .

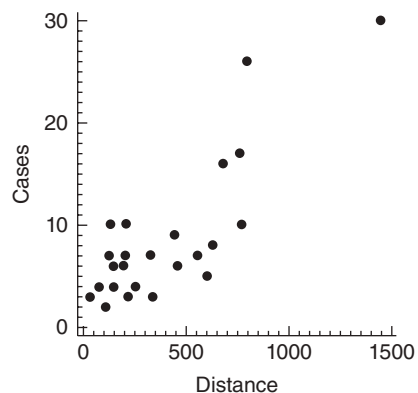


Figure 4.11 Plot of regressor x_1 (cases) versus regressor x_2 (distance) for the delivery time data in Table 3.2.

any problems have occurred. The scatterplot clearly reveals that observation 9 is unusual with respect to both cases and distance ($x_1 = 30$, $x_2 = 1460$); in fact, it is rather remote in x space from the rest of the data. Observation 22 ($x_1 = 26$, $x_2 = 810$) is also quite far from the rest of the data. Points remote in x space can potentially control some of the properties of the regression model. Other formal methods for studying this are discussed in Chapter 6.

The **problem situation** often suggests other types of residual plots. For example, consider the delivery time data in Example 3.1. The 25 observations in Table 3.2 were collected on truck routes in four different cities. Observations 1–7 were collected in San Diego, observations 8–17 in Boston, observations 18–23 in Austin, and observations 24 and 25 in Minneapolis. We might suspect that there is a difference in delivery operations from city to city due to such factors as different types of equipment, different levels of crew training and experience, or motivational factors influenced by management policies. These factors could result in a “site” effect that is not incorporated in the present equation. To investigate this, we plot the residuals by site in Figure 4.12. We see from this plot that there is some imbalance in the distribution of positive and negative residuals at each site. Specifically, there is an apparent tendency for the model to overpredict delivery times in Austin and underpredict delivery times in Boston. This could happen because of the site-dependent factors mentioned above or because one or more important regressors have been omitted from the model.

Statistical Tests on Residuals We may apply statistical tests to the residuals to obtain quantitative measures of some of the model inadequacies discussed above. For example, see Anscombe [1961, 1967], Anscombe and Tukey [1963], Andrews [1971], Looney and Gullledge [1985], Levine [1960], and Cook and Weisberg [1983]. Several formal statistical testing procedures for residuals are discussed in Draper and Smith [1998] and Neter, Kutner, Nachtsheim, and Wasserman [1996].

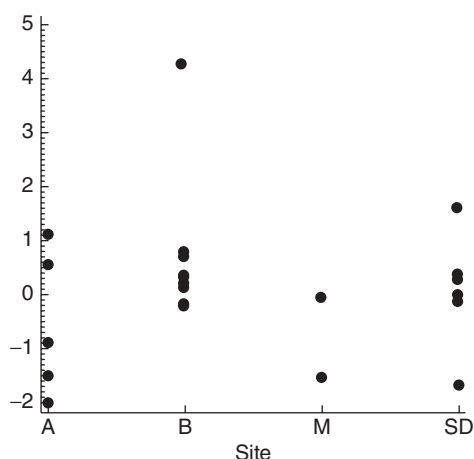


Figure 4.12 Plot of externally studentized residuals by site (city) for the delivery time data in Table 3.2.

In our experience, statistical tests on regression model residuals are not widely used. In most practical situations the residual plots are more informative than the corresponding tests. However, since residual plots do require skill and experience to interpret, the statistical tests may occasionally prove useful. For a good example of the use of statistical tests in conjunction with plots see Feder [1974].

4.3 PRESS STATISTIC

In Section 4.2.2 we defined the PRESS residuals as $e_{(i)} = y_i - \hat{y}_{(i)}$, where $\hat{y}_{(i)}$ is the predicted value of the i th observed response based on a model fit to the remaining $n - 1$ sample points. We noted that large PRESS residuals are potentially useful in identifying observations where the model does not fit the data well or observations for which the model is likely to provide poor future predictions.

Allen [1971, 1974] has suggested using the prediction error sum of squares (or the PRESS statistic), defined as the sum of the squared PRESS residuals, as a measure of model quality. The PRESS statistic is

$$\begin{aligned} \text{PRESS} &= \sum_{i=1}^n [y_i - \hat{y}_{(i)}]^2 \\ &= \sum_{i=1}^n \left(\frac{e_i}{1 - h_{ii}} \right)^2 \end{aligned} \quad (4.17)$$

PRESS is generally regarded as a measure of how well a regression model will perform in **predicting new data**. A model with a small value of PRESS is desired.

Example 4.6 The Delivery Time Data

Column 5 of Table 4.1 shows the calculations of the PRESS residuals for the delivery time data of Example 3.1. Column 7 of Table 4.1 contains the squared PRESS residuals, and the PRESS statistic is shown at the foot of this column. The value of $\text{PRESS} = 457.4000$ is nearly twice as large as the residual sum of squares for this model, $SS_{\text{Res}} = 233.7260$. Notice that almost half of the PRESS statistic is contributed by point 9, a relatively remote point in x space with a moderately large residual. This indicates that the model will not likely predict new observations with large case volumes and long distances particularly well. ■

R^2 for Prediction Based on PRESS The PRESS statistic can be used to compute an R^2 -like statistic for prediction, say

$$R^2_{\text{prediction}} = 1 - \frac{\text{PRESS}}{SS_T} \quad (4.18)$$

This statistic gives some indication of the predictive capability of the regression model. For the soft drink delivery time model we find

$$\begin{aligned}
 R^2_{\text{prediction}} &= 1 - \frac{\text{PRESS}}{SS_T} \\
 &= 1 - \frac{457.4000}{5784.5426} \\
 &= 0.9209
 \end{aligned}$$

Therefore, we could expect this model to “explain” about 92.09% of the variability in predicting new observations, as compared to the approximately 95.96% of the variability in the original data explained by the least-squares fit. The predictive capability of the model seems satisfactory, overall. However, recall that the individual PRESS residuals indicated that observations that are similar to point 9 may not be predicted well.

Using PRESS to Compare Models One very important use of the PRESS statistic is in comparing regression models. Generally, a model with a small value of PRESS is preferable to one where PRESS is large. For example, when we added x_2 = distance to the regression model for the delivery time data containing x_1 = cases, the value of PRESS decreased from 733.55 to 457.40. This is an indication that the two-regressor model is likely to be a better predictor than the model containing only x_1 = cases.

4.4 DETECTION AND TREATMENT OF OUTLIERS

An outlier is an extreme observation; one that is considerably different from the majority of the data. Residuals that are considerably larger in absolute value than the others, say three or four standard deviations from the mean, indicate potential y space outliers. Outliers are data points that are not typical of the rest of the data. Depending on their location in x space, outliers can have moderate to severe effects on the regression model (e.g., see Figures 2.6–2.8). Residual plots against $\hat{y}_i$ and the normal probability plot are helpful in identifying outliers. Examining **scaled residuals**, such as the studentized and R -student residuals, is an excellent way to identify potential outliers. An excellent general treatment of the outlier problems is in Barnett and Lewis [1994]. Also see Myers [1990] for a good discussion.

Outliers should be carefully investigated to see if a reason for their unusual behavior can be found. Sometimes outliers are “bad” values, occurring as a result of unusual but explainable events. Examples include faulty measurement or analysis, incorrect recording of data, and failure of a measuring instrument. If this is the case, then the outlier should be corrected (if possible) or deleted from the data set. Clearly discarding bad values is desirable because least squares pulls the fitted equation toward the outlier as it minimizes the residual sum of squares. However, we emphasize that there should be strong nonstatistical evidence that the outlier is a bad value before it is discarded.

Sometimes we find that the outlier is an unusual but perfectly plausible observation. Deleting these points to “improve the fit of the equation” can be dangerous,

as it can give the user a false sense of precision in estimation or prediction. Occasionally we find that the outlier is more important than the rest of the data because it may control many key model properties. Outliers may also point out inadequacies in the model, such as failure to fit the data well in a certain region of x space. If the outlier is a point of particularly desirable response (e.g., low cost, high yield), knowledge of the regressor values when that response was observed may be extremely valuable. Identification and follow-up analyses of outliers often result in process improvement or new knowledge concerning factors whose effect on the response was previously unknown.

Various statistical tests have been proposed for detecting and rejecting outliers. For example, see Barnett and Lewis [1994]. Stefansky [1971, 1972] has proposed an approximate test for identifying outliers based on the maximum normed residual $|e_i|/\sqrt{\sum_{i=1}^n e_i^2}$ that is particularly easy to apply. Examples of this test and other related references are in Cook and Prescott [1981], Daniel [1976], and Williams [1973]. See also Appendix C.9. While these tests may be useful for identifying outliers, they should not be interpreted to imply that the points so discovered should be automatically rejected. As we have noted, these points may be important clues containing valuable information.

The effect of outliers on the regression model may be easily checked by dropping these points and refitting the regression equation. We may find that the values of the regression coefficients or the summary statistics such as the t or F statistic, R^2 , and the residual mean square may be very sensitive to the outliers. Situations in which a relatively small percentage of the data has a significant impact on the model may not be acceptable to the user of the regression equation. Generally we are happier about assuming that a regression equation is valid if it is not overly sensitive to a few observations. We would like the regression relationship to be embedded in all of the observations and not merely an artifice of a few points.

Example 4.7 The Rocket Propellant Data

Figure 4.13 presents the normal probability plot of the externally studentized residuals and the plot of the externally studentized residuals versus the predicted $\hat{y}_i$ for the rocket propellant data introduced in Example 2.1. We note that there are two large negative residuals that lie quite far from the rest (observations 5 and 6 in Table 2.1). These points are potential outliers. These two points tend to give the normal probability plot the appearance of one for skewed data. Note that observation 5 occurs at a relatively low value of age (5.5 weeks) and observation 6 occurs at a relatively high value of age (19 weeks). Thus, these two points are widely separated in x space and occur near the extreme values of x , and they may be influential in determining model properties. Although neither residual is excessively large, the overall impression from the residual plots (Figure 4.13) is that these two observations are distinctly from the others.

To investigate the influence of these two points on the model, a new regression equation is obtained with observations 5 and 6 deleted. A comparison of the summary statistics from the two models is given below.

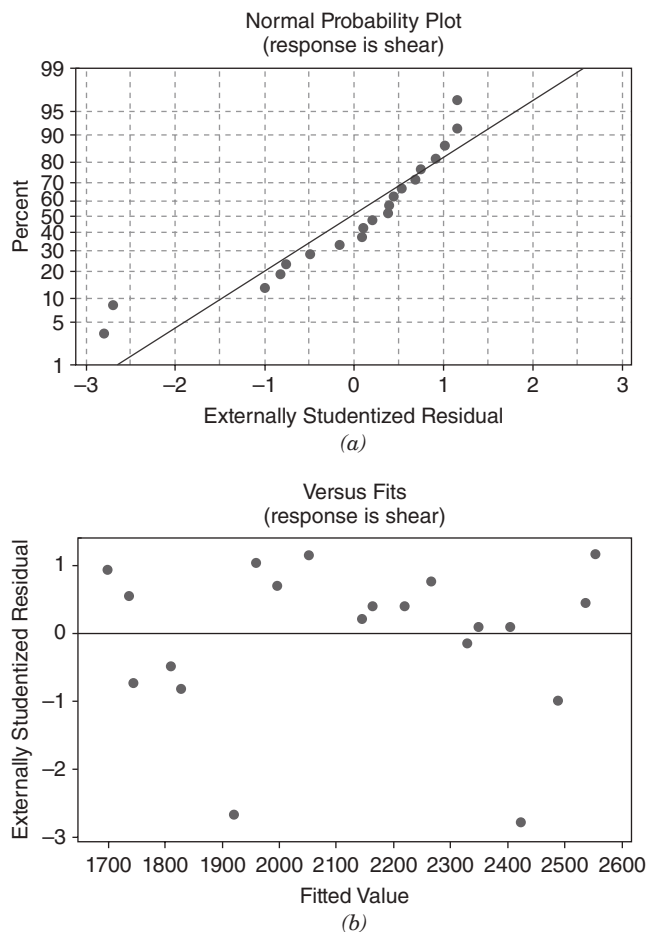


Figure 4.13 Externally studentized residual plots for the rocket propellant data: (a) the normal probability plot; (b) residuals versus predicted $\hat{y}_i$.

	Observations 5 and 6 IN	Observations 5 and 6 OUT
$\hat{\beta}_0$	2627.82	2658.97
$\hat{\beta}_1$	-37.15	-37.69
R^2	0.9018	0.9578
MS_{Res}	9244.59	3964.63
$\text{se}(\hat{\beta}_1)$	2.89	1.98

Deleting points 5 and 6 has almost no effect on the estimates of the regression coefficients. There has, however, been a dramatic reduction in the residual mean square, a moderate increase in R^2 , and approximately a one-third reduction in the standard error of $\hat{\beta}_1$.

Since the estimates of the parameters have not changed dramatically, we conclude that points 5 and 6 are not overly influential. They lie somewhat off the line passing through the other 18 points, but they do not control the slope and intercept. However, these two residuals make up approximately 56% of the residual sum of squares. Thus, if these points are truly bad values and should be deleted, the precision of the parameter estimates would be improved and the widths of confidence and prediction intervals could be substantially decreased.

Figure 4.14 shows the normal probability plot of the externally studentized residuals and the plot of the externally studentized residuals versus $\hat{y}_i$ for the model with points 5 and 6 deleted. These plots do not indicate any serious departures from assumptions.

Further examination of points 5 and 6 fails to reveal any reason for the unusually low propellant shear strengths obtained. Therefore, we should not discard these two points. However, we feel relatively confident that including them does not seriously limit the use of the model. ■

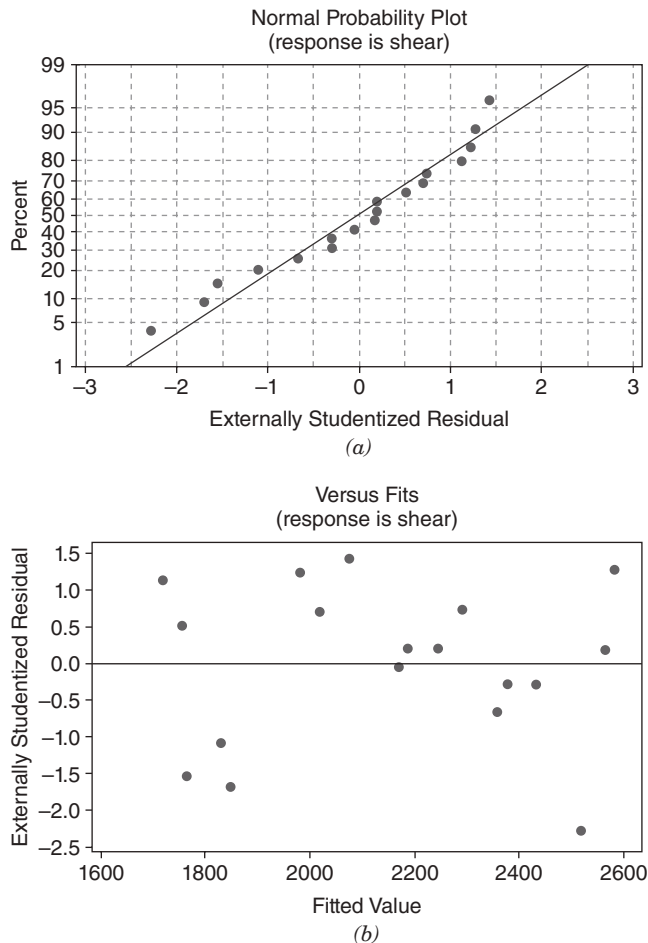


Figure 4.14 Residual plots for the rocket propellant data with observations 5 and 6 removed: (a) the normal probability plot; (b) residuals versus predicted $\hat{y}_i$.

4.5 LACK OF FIT OF THE REGRESSION MODEL

A famous quote attributed to George Box is “All models are wrong; some models are useful.” This comment goes to heart of why tests for lack-of-fit are important. In basic English, lack-of-fit is “the terms that we could have fit to the model but chose not to fit.” For example, only two distinct points are required to fit a straight line. If we have three distinct points, then we could fit a parabola (a second-order model). If we choose to fit only the straight line, then we note that in general the straight line does not go through all three points. We typically assume that this phenomenon is due to error. On the other hand, the true underlying mechanism could really be quadratic. In the process, what we claim to be random error is actually a systematic departure as the result of not fitting enough terms. In the simple linear regression context, if we have n distinct data points, we can always fit a polynomial of order up to $n - 1$. When we choose to fit a straight line, we give up $n - 2$ degrees of freedom to estimate the error term when we could have chosen to fit these other higher-order terms.

4.5.1 A Formal Test for Lack of Fit

The formal statistical test for the lack of fit of a regression model assumes that the normality, independence, and constant-variance requirements are met and that only the first-order or straight-line character of the relationship is in doubt. For example, consider the data in Figure 4.15. There is some indication that the straight-line fit is not very satisfactory. Perhaps, a quadratic term (x^2) should be added, or perhaps another regressor should be added. It would be helpful to have a test procedure to determine if systematic lack of fit is present.

The lack-of-fit test requires that we have replicate observations on the response y for at least one level of x . We emphasize that these should be true replications, not just duplicate readings or measurements of y . For example, suppose that y is product viscosity and x is temperature. True replication consists of running n_i separate experiments at $x = x_i$ and observing viscosity, not just running a single

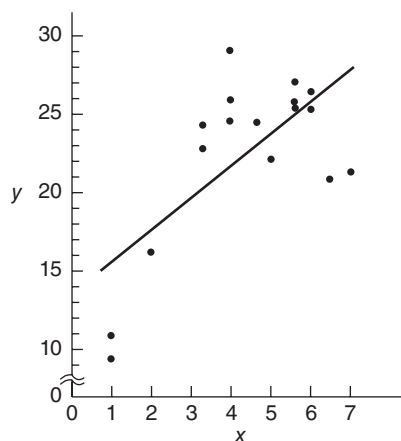


Figure 4.15 Data illustrating lack of fit of the straight-line model.

experiment at x_i and measuring viscosity n_i times. The readings obtained from the latter procedure provide information only on the variability of the method of measuring viscosity. The error variance σ^2 includes this measurement error and the variability associated with reaching and maintaining the same temperature level in different experiments. These replicated observations are used to obtain a **model-independent estimate** of σ^2 .

Suppose that we have n_i observations on the response at the i th level of the regressor x_i , $i = 1, 2, \dots, m$. Let y_{ij} denote the j th observation on the response at x_i , $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n_i$. There are $n = \sum_{i=1}^m n_i$ total observations. The test procedure involves partitioning the residual sum of squares into two components, say

$$SS_{\text{Res}} = SS_{\text{PE}} + SS_{\text{LOF}}$$

where SS_{PE} is the sum of squares due to **pure error** and SS_{LOF} is the sum of squares due to **lack of fit**.

To develop this partitioning of SS_{Res} , note that the (ij) th residual is

$$y_{ij} - \hat{y}_i = (y_{ij} - \bar{y}_i) + (\bar{y}_i - \hat{y}_i) \quad (4.19)$$

where $\bar{y}_i$ is the average of the n_i observations at x_i . Squaring both sides of Eq. (4.19) and summing over i and j yields

$$\sum_{i=1}^m \sum_{j=1}^{n_i} (y_{ij} - \hat{y}_i)^2 = \sum_{i=1}^m \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2 + \sum_{i=1}^m n_i (\bar{y}_i - \hat{y}_i)^2 \quad (4.20)$$

since the cross-product term equals zero.

The left-hand side of Eq. (4.20) is the usual residual sum of squares. The two components on the right-hand side measure pure error and lack of fit. We see that the pure-error sum of squares

$$SS_{\text{PE}} = \sum_{i=1}^m \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2 \quad (4.21)$$

is obtained by computing the corrected sum of squares of the repeat observations at each level of x and then pooling over the m levels of x . If the assumption of constant variance is satisfied, this is a **model-independent measure of pure error** since only the variability of the y 's at each x level is used to compute SS_{PE} . Since there are $n_i - 1$ degrees of freedom for pure error at each level x_i , the total number of degrees of freedom associated with the pure-error sum of squares is

$$\sum_{i=1}^m (n_i - 1) = n - m \quad (4.22)$$

The sum of squares for lack of fit

$$SS_{\text{LOF}} = \sum_{i=1}^m n_i (\bar{y}_i - \hat{y}_i)^2 \quad (4.23)$$

is a weighted sum of squared deviations between the mean response $\bar{y}_i$ at each x level and the corresponding fitted value. If the fitted values $\hat{y}_i$ are close to the corresponding average responses $\bar{y}_i$, then there is a strong indication that the regression function is linear. If the $\hat{y}_i$ deviate greatly from the $\bar{y}_i$, then it is likely that the regression function is not linear. There are $m - 2$ degrees of freedom associated with SS_{LOF} , since there are m levels of x and two degrees of freedom are lost because two parameters must be estimated to obtain the $\bar{y}_i$. Computationally we usually obtain SS_{LOF} by subtracting SS_{PE} from SS_{Res} .

The test statistic for lack of fit is

$$F_0 = \frac{SS_{\text{LOF}} / (m - 2)}{SS_{\text{PE}} / (n - m)} = \frac{MS_{\text{LOF}}}{MS_{\text{PE}}} \quad (4.24)$$

The expected value of MS_{PE} is σ^2 , and the expected value of MS_{LOF} is

$$E(MS_{\text{LOF}}) = \sigma^2 + \frac{\sum_{i=1}^m n_i [E(y_i) - \beta_0 - \beta_1 x_i]^2}{m - 2} \quad (4.25)$$

If the true regression function is linear, then $E(y_i) = \beta_0 + \beta_1 x_i$, and the second term of Eq. (4.25) is zero, resulting in $E(MS_{\text{LOF}}) = \sigma^2$. However, if the true regression function is not linear, then $E(y_i) \neq \beta_0 + \beta_1 x_i$, and $E(MS_{\text{LOF}}) > \sigma^2$. Furthermore, if the true regression function is linear, then the statistic F_0 follows the $F_{m-2, n-m}$ distribution. Therefore, to test for lack of fit, we would compute the test statistic F_0 and conclude that the regression function is not linear if $F_0 > F_{\alpha, m-2, n-m}$.

This test procedure may be easily introduced into the analysis of variance conducted for significance of regression. If we conclude that the regression function is not linear, then the tentative model must be abandoned and attempts made to find a more appropriate equation. Alternatively, if F_0 does not exceed $F_{\alpha, m-2, n-m}$, there is no strong evidence of lack of fit, and MS_{PE} and MS_{LOF} are often combined to estimate σ^2 .

Ideally, we find that the F ratio for lack of fit is not significant, and the hypothesis of significance of regression ($H_0: \beta_1 = 0$) is rejected. Unfortunately, this does not guarantee that the model will be satisfactory as a prediction equation. Unless the variation of the predicted values is large relative to the random error, the model is not estimated with sufficient precision to yield satisfactory predictions. That is, the model may have been fitted to the errors only. Some analytical work has been done on developing criteria for judging the adequacy of the regression model from a prediction point of view. See Box and Wetz [1973], Ellerton [1978], Gunst and Mason [1979], Hill, Judge, and Fomby [1978], and Suich and Derringer [1977]. The Box and Wetz work suggests that the observed F ratio must be at least four or five times the critical value from the F table if the regression model is to be useful as a predictor, that is, if the spread of predicted values is to be large relative to the noise.

A relatively simple measure of potential prediction performance is found by comparing the range of the fitted values $\hat{y}_i$ (i.e., $\hat{y}_{\max} - \hat{y}_{\min}$) to their average standard error. It can be shown that, regardless of the form of the model, the average variance of the fitted values is

$$\overline{\text{Var}(\hat{y})} = \frac{1}{n} \sum_{i=1}^n \text{Var}(\hat{y}_i) = \frac{p\sigma^2}{n} \quad (4.26)$$

where p is the number of parameters in the model. In general, the model is not likely to be a satisfactory predictor unless the range of the fitted values $\hat{y}_i$ is large relative to their average estimated standard error $\sqrt{(p\hat{\sigma}^2)/n}$, where $\hat{\sigma}^2$ is a model-independent estimate of the error variance.

Example 4.8 Testing for Lack of Fit

The data from Figure 4.15 are shown below:

x	1.0	1.0	2.0	3.3	3.3	4.0	4.0	4.0	4.7	5.0
y	10.84	9.30	16.35	22.88	24.35	24.56	25.86	29.16	24.59	22.25
x	5.6	5.6	5.6	6.0	6.0	6.5	6.9			
y	25.90	27.20	25.61	25.45	26.56	21.03	21.46			

The straight-line fit is $\hat{y} = 13.301 + 2.108x$, with $SS_T = 487.6126$, $SS_R = 234.7087$, and $SS_{\text{Res}} = 252.9039$. Note that there are 10 distinct levels of x , with repeat points at $x = 1.0$, $x = 3.3$, $x = 4.0$, $x = 5.6$, and $x = 6.0$. The pure-error sum of squares is computed using the repeat points as follows:

Level of x	$\sum_j (y_{ij} - \bar{y}_i)^2$	Degrees of Freedom
1.0	1.1858	1
3.3	1.0805	1
4.0	11.2467	2
5.6	1.4341	2
6.0	0.6161	1
Total	15.5632	7

The lack-of-fit sum of squares is found by subtraction as

$$\begin{aligned} SS_{\text{LOF}} &= SS_{\text{Res}} - SS_{\text{PE}} \\ &= 252.9039 - 15.5632 = 237.3407 \end{aligned}$$

with $m - 2 = 10 - 2 = 8$ degrees of freedom. The analysis of variance incorporating the lack-of-fit test is shown in Table 4.4. The lack-of-fit test statistic is $F_0 = 13.34$, and since the P value is very small, we reject the hypothesis that the tentative model adequately describes the data. ■

TABLE 4.4 Analysis of Variance for Example 4.8

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P Value
Regression	234.7087	1	234.7087		
Residual	252.9039	15	16.8603		
(Lack of fit)	237.3407	8	29.6676	13.34	0.0013
(Pure error)	15.5632	7	2.2233		
Total	487.6126	16			

Example 4.9 Testing for Lack of Fit in JMP

Some software packages will perform the lack of fit test automatically if there are replicate observations in the data. In the patient satisfaction data of Appendix Table B.17 there are replicate observations in the severity predictor (they occur at 30, 31, 38, 42, 28, and 50). Figure 4.16 is a portion of the JMP output that results from fitting a simple linear regression model to these data. The F -test for lack of fit in Equation 4.24 is shown in the output. The P -value is 0.0874, so there is some mild indication of lack of fit. Recall from Section 3.6 that when we added the second predictor (age) to this model the quality of the overall fit improved considerably. As this example illustrates, sometimes lack of fit is caused by missing regressors; it isn't always necessary to add higher-order terms to the model. ■

4.5.2 Estimation of Pure Error from Near Neighbors

In Section 4.5.1 we described a test for lack of fit for the linear regression model. The procedure involved partitioning the error or residual sum of squares into a component due to “pure” error and a component due to lack of fit:

$$SS_{\text{Res}} = SS_{\text{PE}} + SS_{\text{LOF}}$$

The pure-error sum of squares SS_{PE} is computed using responses at repeat observations at the same level of x . This is a **model-independent estimate** of σ^2 .

This general procedure can in principle be applied to any regression model. The calculation of SS_{PE} requires repeat observations on the response y at the same set of levels on the regressor variables $x_1, x_2, \dots, x_k$. That is, some of the **rows** of the **X** matrix must be the same. However, repeat observations do not often occur in multiple regression, and the procedure described in Section 4.5.1 is not often useful.

Daniel and Wood [1980] and Joglekar, Schuenemeyer, and La Riccia [1989] have investigated methods for obtaining a model-independent estimate of error when there are no exact repeat points. These procedures search for points in x space that are **near neighbors**, that is, sets of observations that have been taken with nearly

Response Satisfaction Whole Model

Summary of Fit

RSquare	0.426596
RSquare Adj	0.401666
Root Mean Square Error	16.43242
Mean of Response	66.72
Observations (or Sum Wgts)	25

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	4620.482	4620.48	17.1114
Error	23	6210.558	270.02	Prob > F
C. Total	24	10831.040		0.0004*

Lack Of Fit

Source	DF	Sum of Squares	Mean Square	F Ratio
Lack Of Fit	16	5366.0584	335.379	2.7799
Pure Error	7	844.5000	120.643	Prob > F
Total Error	23	6210.5584		0.0874
				Max RSq
				0.9220

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob> t
Intercept	115.6239	12.27059	9.42	<.0001*
Severity	-1.06498	0.257454	-4.14	0.0004*

Figure 4.16 JMP output for the simple linear regression model relating satisfaction to severity.

identical levels of $x_1, x_2, \dots, x_k$. The responses y_i from such near neighbors can be considered as repeat points and used to obtain an estimate of pure error. As a measure of the distance between any two points, for example, $x_{i1}, x_{i2}, \dots, x_{ik}$ and $x_{i'1}, x_{i'2}, \dots, x_{i'k}$, we will use the weighted sum of squared distance (WSSD)

$$D_{ii'}^2 = \sum_{j=1}^k \left[\frac{\hat{\beta}_j (x_{ij} - x_{i'j})}{\sqrt{MS_{\text{Res}}}} \right]^2 \quad (4.27)$$

Pairs of points that have small values of $D_{ii'}^2$ are “near neighbors,” that is, they are relatively close together in x space. Pairs of points for which $D_{ii'}^2$ is large (e.g., $D_{ii'}^2 \gg 1$) are widely separated in x space. The residuals at two points with a small value of $D_{ii'}^2$ can be used to obtain an estimate of pure error. The estimate is obtained from the range of the residuals at the points i and i' , say

$$E_i = |e_i - e_{i'}|$$

There is a relationship between the range of a sample from a normal population and the population standard deviation. For samples of size 2, this relationship is

$$\hat{\sigma} = (1.128)^{-1} E = 0.886E$$

The quantity $\hat{\sigma}$ so obtained is an estimate of the standard deviation of pure error.

An efficient algorithm may be used to compute this estimate. A computer program for this algorithm is given in Montgomery, Martin, and Peck [1980]. First arrange the data points $x_{i1}, x_{i2}, \dots, x_{ik}$ in order of increasing $\hat{y}_i$. Note that points with very different values of $\hat{y}_i$ cannot be near neighbors, but those with similar values of $\hat{y}_i$ could be neighbors (or they could be near the same contour of constant $\hat{y}$ but far apart in some x coordinates). Then:

1. Compute the values of $D_{ii'}^2$ for all $n - 1$ pairs of points with adjacent values of $\hat{y}$. Repeat this calculation for the pairs of points separated by one, two, and three intermediate $\hat{y}$ values. This will produce $4n - 10$ values of $D_{ii'}^2$.
2. Arrange the $4n - 10$ values of $D_{ii'}^2$ found in 1 above in ascending order. Let E_u , $u = 1, 2, \dots, 4n - 10$, be the range of the residuals at these points.
3. For the first m values of E_u , calculate an estimate of the standard deviation of pure error as

$$\hat{\sigma} = \frac{0.886}{m} \sum_{u=1}^m E_u \quad (4.28)$$

Note that $\hat{\sigma}$ is based on the average range of the residuals associated with the m smallest values of $D_{ii'}^2$; m must be chosen after inspecting the values of $D_{ii'}^2$. One should not include values of E_u in the calculations for which the weighted sum of squared distance is too large.

Example 4.10 The Delivery Time Data

We use the procedure described above to calculate an estimate of the standard deviation of pure error for the soft drink delivery time data from Example 3.1. Table 4.5 displays the calculation of $D_{ii'}^2$ for pairs of points that, in terms of $\hat{y}$, are adjacent, one apart, two apart, and three apart. The R columns in this table identify the 15 smallest values of $D_{ii'}^2$. The residuals at these 15 pairs of points are used to estimate σ . These calculations yield $\hat{\sigma} = 1.969$ and are summarized in Table 4.6. From Table 3.4, we find that $\sqrt{MS_{\text{Res}}} = \sqrt{10.6239} = 3.259$. Now if there is no appreciable lack of fit, we would expect to find that $\hat{\sigma} = \sqrt{MS_{\text{Res}}}$. In this case $\sqrt{MS_{\text{Res}}}$ is about 65% larger than $\hat{\sigma}$, indicating some lack of fit. This could be due to the effects of regressors not presently in the model or the presence of one or more outliers. ■

TABLE 4.5 Calculation of D_H^2 for Example 4.10

Near-Neighbor Calculations Delta Residuals and the Weighted Standardized Squared Distances of Near Neighbors														
Adjacent					1 Apart					2 Apart				
Observation	Ordered Fitted, y	Residual	Delta	D_H^2	R ^a	Delta	D_H^2	R	Delta	D_H^2	R	Delta	D_H^2	R
7	7.155	.845	.949	.3524E+00		4.080	.1001E+01		.302	.4814E+00		1.057	.1014E+01	
19	7.707	1.793	3.131	.2835E+00	12	.647	.6594E+00		2.006	.4989E+00		1.843	.1800E+01	
4	9.956	4.924	3.778	.6275E+00		5.137	.9544E-01	3	4.974	.1562E+01		3.897	.5965E+00	
2	10.354	1.146	1.359	.3412E+00	15	1.196	.2805E+00	11	1.119	.2696E+00	9	1.591	.2307E+01	
25	10.963	-.213	.163	.9489E+00		1.240	.2147E+00	6	.232	.9831E+00		.649	.1032E+01	
3	12.080	-.050	1.077	.3865E+00		.395	.2915E+01		.486	.2594E+01		3.498	.4775E+01	
13	12.473	1.027	1.471	.1198E+01		.591	.1042E+01		2.422	.2507E+01		.130	.2251E+01	
5	14.194	-.444	.881	.4869E-01	2	3.893	.2521E+00	8	1.601	.3159E+00	13	.155	.8768E+00	
17	14.914	.436	3.012	.3358E+00	14	.720	.2477E+00	7	.726	.5749E+00		.631	.1337E+01	
18	15.551	3.449	2.292	.1185E+00	5	3.738	.7636E+00		2.381	.2367E+01		1.072	.5341E+01	
8	16.673	1.157	1.446	.2805E+00	10	.089	.1483E+01		1.220	.4022E+01		3.771	.2307E+01	
6	18.400	-.290	1.357	.5851E+00		2.666	.2456E+01		2.325	.2915E+01		.303	.2470E+01	
14	18.682	1.068	1.309	.6441E+00		3.682	.5952E+01		1.661	.5121E+01		6.096	.4328E+00	
10	19.124	2.376	4.991	.1036E+02		2.969	.9107E+01		7.404	.1023E+01		1.705	.4412E+01	
21	20.514	-2.614	2.021	.1096E+00	4	2.414	.5648E+01		3.285	.2093E+01		1.993	.2117E+01	
12	21.593	-.593	4.435	.4530E+01		1.264	.1303E+01		4.015	.1321E+01		3.980	.4419E+01	
1	21.708	-5.028	5.699	.1227E+01		.421	.1219E+01		.455	.3553E+00		4.365	.3121E+01	
15	23.329	.671	5.279	.7791E-04	1	5.244	.9269E+00		1.334	.2341E+01		1.566	.1316E+02	
23	23.358	-4.608	.035	.9124E+00		3.945	.2316E+01		6.845	.1315E+02		1.180	.1772E+02	
24	24.403	-4.573	3.910	.1370E+01		6.810	.1578E+02		1.215	.2026E+02		.886	.8023E+02	
16	29.663	-.663	2.900	.8999E+01		5.125	.1204E+02		3.024	.6294E+02		8.083	.1074E+03	
11	38.093	2.237	8.025	.3767E+00		5.924	.2487E+02		5.182	.5978E+02				
20	40.888	-5.788	2.101	.1994E+02		13.208	.5081E+02							
22	56.007	-3.687	11.106	.1216E+02										
9	71.820	7.420												

^a Column R gives the rank order of the 15 smallest D_H^2 values.

TABLE 4.6 Calculation of $\hat{\sigma}$ for Example 4.10

Number	Standard Deviation Estimated from Residuals of Neighboring Observations				
	Cumulative Standard Deviation	Ordered by D_{ii}^2			
		D_{ii}^2	Observation	Observation	Delta Residual
1	.4677E + 01	.7791E - 04	15	23	5.2788
2	.2729E + 01	.4859E - 01	5	17	0.8807
3	.3336E + 01	.9544E - 01	4	25	5.1369
4	.2950E + 01	.1096E + 00	21	12	2.0211
5	.2766E + 01	.1185E + 00	18	8	2.2920
6	.2488E + 01	.2147E + 00	25	13	1.2396
7	.2224E + 01	.2477E + 00	17	8	0.7203
8	.2377E + 01	.2521E + 00	5	18	3.8930
9	.2125E + 01	.2696E + 00	2	13	0.1194
10	.2040E + 01	.2805E + 00	8	6	1.4462
11	.1951E + 01	.2805E + 00	2	3	1.1962
12	.2020E + 01	.2835E + 00	19	4	3.1312
13	.1973E + 01	.3159E + 00	5	8	1.6010
14	.2023E + 01	.3358E + 00	17	18	3.0123
15	.1969E + 01	.3412E + 00	2	25	1.3590
16	.1898E + 01	.3524E + 00	7	19	0.9486
17	.1810E + 01	.3553E + 00	1	24	0.4552
18	.2105E + 01	.3767E + 00	11	20	8.0255
19	.2044E + 01	.3865E + 00	3	13	1.0768
20	.2212E + 01	.4328E + 00	14	1	6.0956
21	.2119E + 01	.4814E + 00	7	2	0.3018
22	.2104E + 01	.4989E + 00	19	25	2.0058
23	.2040E + 01	.5749E + 00	17	6	0.7259
24	.2005E + 01	.5851E + 00	6	14	1.3571
25	.2063E + 01	.5965E + 00	4	13	3.8973
26	.2113E + 01	.6275E + 00	4	2	3.7780
27	.2077E + 01	.6441E + 00	14	10	1.3089
28	.2024E + 01	.6594E + 00	19	2	0.6468
29	.2068E + 01	.7636E + 00	18	6	3.7382
30	.2004E + 01	.8768E + 00	5	6	0.1548
31	.1940E + 01	.9124E + 00	23	24	0.0347
32	.2025E + 01	.9269E + 00	15	24	5.2441
33	.1968E + 01	.9489E + 00	25	3	0.1628
34	.1916E + 01	.9831E + 00	25	5	0.2318
35	.1964E + 01	.1001E + 01	7	4	4.0797
36	.1936E + 01	.1014E + 01	7	25	1.0572
37	.2061E + 01	.1023E + 01	10	1	7.4045
38	.2022E + 01	.1032E + 01	25	17	0.6489
39	.1983E + 01	.1042E + 01	13	17	0.5907
40	.1966E + 01	.1198E + 01	13	5	1.4714

PROBLEMS

- 4.1** Consider the simple regression model fit to the National Football League team performance data in Problem 2.1.
- Construct a normal probability plot of the residuals. Does there seem to be any problem with the normality assumption?
 - Construct and interpret a plot of the residuals versus the predicted response.
 - Plot the residuals versus the team passing yardage, x_2 . Does this plot indicate that the model will be improved by adding x_2 to the model?
- 4.2** Consider the multiple regression model fit to the National Football League team performance data in Problem 3.1.
- Construct a normal probability plot of the residuals. Does there seem to be any problem with the normality assumption?
 - Construct and interpret a plot of the residuals versus the predicted response.
 - Construct plots of the residuals versus each of the regressor variables. Do these plots imply that the regressor is correctly specified?
 - Construct the partial regression plots for this model. Compare the plots with the plots of residuals versus regressors from part c above. Discuss the type of information provided by these plots.
 - Compute the studentized residuals and the R -student residuals for this model. What information is conveyed by these scaled residuals?
- 4.3** Consider the simple linear regression model fit to the solar energy data in Problem 2.3.
- Construct a normal probability plot of the residuals. Does there seem to be any problem with the normality assumption?
 - Construct and interpret a plot of the residuals versus the predicted response.
- 4.4** Consider the multiple regression model fit to the gasoline mileage data in Problem 3.5.
- Construct a normal probability plot of the residuals. Does there seem to be any problem with the normality assumption?
 - Construct and interpret a plot of the residuals versus the predicted response.
 - Construct and interpret the partial regression plots for this model.
 - Compute the studentized residuals and the R -student residuals for this model. What information is conveyed by these scaled residuals?
- 4.5** Consider the multiple regression model fit to the house price data in Problem 3.7.
- Construct a normal probability plot of the residuals. Does there seem to be any problem with the normality assumption?
 - Construct and interpret a plot of the residuals versus the predicted response.
 - Construct the partial regression plots for this model. Does it seem that some variables currently in the model are not necessary?
 - Compute the studentized residuals and the R -student residuals for this model. What information is conveyed by these scaled residuals?

- 4.6** Consider the simple linear regression model fit to the oxygen purity data in Problem 2.7.
- Construct a normal probability plot of the residuals. Does there seem to be any problem with the normality assumption?
 - Construct and interpret a plot of the residuals versus the predicted response.
- 4.7** Consider the simple linear regression model fit to the weight and blood pressure data in Problem 2.10.
- Construct a normal probability plot of the residuals. Does there seem to be any problem with the normality assumption?
 - Construct and interpret a plot of the residuals versus the predicted response.
 - Suppose that the data were collected in the order shown in the table. Plot the residuals versus time order and comment on the plot.
- 4.8** Consider the simple linear regression model fit to the steam plant data in Problem 2.12.
- Construct a normal probability plot of the residuals. Does there seem to be any problem with the normality assumption?
 - Construct and interpret a plot of the residuals versus the predicted response.
 - Suppose that the data were collected in the order shown in the table. Plot the residuals versus time order and comment on the plot.
- 4.9** Consider the simple linear regression model fit to the ozone data in Problem 2.13.
- Construct a normal probability plot of the residuals. Does there seem to be any problem with the normality assumption?
 - Construct and interpret a plot of the residuals versus the predicted response.
 - Plot the residuals versus time order and comment on the plot.
- 4.10** Consider the simple linear regression model fit to the copolyester viscosity data in Problem 2.14.
- Construct a normal probability plot of the unscaled residuals. Does there seem to be any problem with the normality assumption?
 - Repeat part a using the studentized residuals. Is there any substantial difference in the two plots?
 - Construct and interpret a plot of the residuals versus the predicted response.
- 4.11** Consider the simple linear regression model fit to the toluene–tetralin viscosity data in Problem 2.15.
- Construct a normal probability plot of the residuals. Does there seem to be any problem with the normality assumption?
 - Construct and interpret a plot of the residuals versus the predicted response.
- 4.12** Consider the simple linear regression model fit to the tank pressure and volume data in Problem 2.16.
- Construct a normal probability plot of the residuals. Does there seem to be any problem with the normality assumption?

- b. Construct and interpret a plot of the residuals versus the predicted response.
 - c. Suppose that the data were collected in the order shown in the table. Plot the residuals versus time order and comment on the plot.
- 4.13** Problem 3.8 asked you to fit two different models to the chemical process data in Table B.5. Perform appropriate residual analyses for both models. Discuss the results of these analyses. Calculate the PRESS statistic for both models. Do the residual plots and PRESS provide any insight regarding the best choice of model for the data?
- 4.14** Problems 2.4 and 3.5 asked you to fit two different models to the gasoline mileage data in Table B.3. Calculate the PRESS statistic for these two models. Based on this statistic, which model is most likely to provide better predictions of new data?
- 4.15** In Problem 3.9, you were asked to fit a model to the tube-flow reactor data in Table B.6.
- a. Construct a normal probability plot of the residuals. Does there seem to be any problem with the normality assumption?
 - b. Construct and interpret a plot of the residuals versus the predicted response.
 - c. Construct the partial regression plots for this model. Does it seem that some variables currently in the model are not necessary?
- 4.16** In Problem 3.12, you were asked to fit a model to the clathrate formation data in Table B.8.
- a. Construct a normality plot of the residuals from the full model. Does there seem to be any problem with the normality assumption?
 - b. Construct and interpret a plot of the residuals versus the predicted response.
 - c. In Problem 3.12, you were asked to fit a second model. Compute the PRESS statistic for both models. Based on this statistic, which model is most likely to provide better predictions of new data?
- 4.17** In Problem 3.14, you were asked to fit a model to the kinematic viscosity data in Table B.10.
- a. Construct a normality plot of the residuals from the full model. Does there seem to be any problem with the normality assumption?
 - b. Construct and interpret a plot of the residuals versus the predicted response.
 - c. In Problem 3.14, you were asked to fit a second model. Compute the PRESS statistic for both models. Based on this statistic, which model is most likely to provide better predictions of new data?
- 4.18** Coteron, Sanchez, Martinez, and Aracil ("Optimization of the Synthesis of an Analogue of Jojoba Oil Using a Fully Central Composite Design," *Canadian Journal of Chemical Engineering*, 1993) studied the relationship of reaction temperature x_1 , initial amount of catalyst x_2 , and pressure x_3 on the yield of a synthetic analogue to jojoba oil. The following table summarizes the experimental results.

x_1	x_2	x_3	y
-1	-1	-1	17
1	-1	-1	44
-1	1	-1	19
1	1	-1	46
-1	-1	1	7
1	-1	1	55
-1	1	1	15
1	1	1	41
0	0	0	29
0	0	0	28.5
0	0	0	30
0	0	0	27
0	0	0	28

- a. Perform a thorough analysis of the results including residual plots.
- b. Perform the appropriate test for lack of fit.

4.19 Derringer and Suich (“Simultaneous Optimization of Several Response Variables,” *Journal of Quality Technology*, 1980) studied the relationship of an abrasion index for a tire tread compound in terms of three factors: x_1 , hydrated silica level; x_2 , silane coupling agent level; and x_3 , sulfur level. The following table gives the actual results.

x_1	x_2	x_3	y
-1	-1	1	102
1	-1	-1	120
-1	1	-1	117
1	1	1	198
-1	-1	-1	103
1	-1	1	132
-1	1	1	132
1	1	-1	139
0	0	0	133
0	0	0	133
0	0	0	140
0	0	0	142
0	0	0	145
0	0	0	142

- a. Perform a thorough analysis of the results including residual plots.
- b. Perform the appropriate test for lack of fit.

4.20 Myers Montgomery and Anderson-Cook (*Response Surface Methodology 3rd edition*, Wiley, New York, 2009) discuss an experiment to determine the influence of five factors:

x_1 —acid bath temperature
 x_2 —cascade acid concentration
 x_3 —water temperature
 x_4 —sulfide concentration
 x_5 —amount of chlorine bleach

on an appropriate measure of the whiteness of rayon (y). The engineers conducting this experiment wish to minimize this measure. The experimental results follow.

Acid Temp.	Acid Conc.	Water Temp.	Sulfide Conc.	Amount of Bleach	y
35	0.3	82	0.2	0.3	76.5
35	0.3	82	0.3	0.5	76.0
35	0.3	88	0.2	0.5	79.9
35	0.3	88	0.3	0.3	83.5
35	0.7	82	0.2	0.5	89.5
35	0.7	82	0.3	0.3	84.2
35	0.7	88	0.2	0.3	85.7
35	0.7	88	0.3	0.5	99.5
55	0.3	82	0.2	0.5	89.4
55	0.3	82	0.3	0.3	97.5
55	0.3	88	0.2	0.3	103.2
55	0.3	88	0.3	0.5	108.7
55	0.7	82	0.2	0.3	115.2
55	0.7	82	0.3	0.5	111.5
55	0.7	88	0.2	0.5	102.3
55	0.7	88	0.3	0.3	108.1
25	0.5	85	0.25	0.4	80.2
65	0.5	85	0.25	0.4	89.1
45	0.1	85	0.25	0.4	77.2
45	0.9	85	0.25	0.4	85.1
45	0.5	79	0.25	0.4	71.5
45	0.5	91	0.25	0.4	84.5
45	0.5	85	0.15	0.4	77.5
45	0.5	85	0.35	0.4	79.2
45	0.5	85	0.25	0.2	71.0
45	0.5	85	0.25	0.6	90.2

- a. Perform a thorough analysis of the results including residual plots.
- b. Perform the appropriate test for lack of fit.

4.21 Consider the test for lack of fit. Find $E(MS_{PE})$ and $E(MS_{LOF})$.

4.22 Table B.14 contains data on the transient points of an electronic inverter. Using only the regressors $x_1, \dots, x_4$, fit a multiple regression model to these data.

- a. Investigate the adequacy of the model.
 - b. Suppose that observation 2 was recorded incorrectly. Delete this observation, refit the model, and perform a thorough residual analysis. Comment on the difference in results that you observe.
- 4.23** Consider the advertising data given in Problem 2.18.
- a. Construct a normal probability plot of the residuals from the full model. Does there seem to be any problem with the normality assumption?
 - b. Construct and interpret a plot of the residuals versus the predicted response.
- 4.24** Consider the air pollution and mortality data given in Problem 3.15 and Table B.15.
- a. Construct a normal probability plot of the residuals from the full model. Does there seem to be any problem with normality assumption?
 - b. Construct and interpret a plot of the residuals versus the predicted response.
- 4.25** Consider the life expectancy data given in Problem 3.16 and Table B.16.
- a. For each model construct a normal probability plot of the residuals from the full model. Does there seem to be any problem with the normality assumption?
 - b. For each model construct and interpret a plot of the residuals versus the predicted response.
- 4.26** Consider the multiple regression model for the patient satisfaction data in Section 3.6. Analyse the residuals from this model and comment on model adequacy.
- 4.27** Consider the fuel consumption data in Table B.18. For the purposes of this exercise, ignore regressor x_1 . Perform a thorough residual analysis of these data. What conclusions do you draw from this analysis?
- 4.28** Consider the wine quality of young red wines data in Table B.19. For the purposes of this exercise, ignore regressor x_1 . Perform a thorough residual analysis of these data. What conclusions do you draw from this analysis?
- 4.29** Consider the methanol oxidation data in Table B.20. Perform a thorough analysis of these data. What conclusions do you draw from this residual analysis?